Data Collector Support Matrix

Cloud Insights

NetApp
March 27, 2024

This PDF was generated from https://docs.netapp.com/us-en/cloudinsights/reference_data_collector_support_matrix.html on March 27, 2024. Always check docs.netapp.com for the latest.

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Cloud Insights Data Collector Support Matrix

The Data Collector Support Matrix provides reference for Data Collectors supported by Cloud Insights, including vendor and model information.

HP Enterprise 3PAR / Alletra 9000 / Primera StoreServ Storage

Models and versions supported by this data collector:

Models	Firmware versions
HPE Alletra 9080 HPE_3PAR 20450 HPE_3PAR 20800 HPE_3PAR 20850 HPE_3PAR 20850_R2 HPE_3PAR 7200c HPE_3PAR 7400 HPE_3PAR 7440c HPE_3PAR 7450c HPE_3PAR 8200 HPE_3PAR 8400 HPE_3PAR 8440 HPE_3PAR 8450 HPE_3PAR 9450 HPE_3PAR A630 HPE_3PAR A650 HPE_3PAR A670 HP_3PAR 20800 HP_3PAR 7200 HP_3PAR 7200c HP_3PAR 7400 HP_3PAR 7400c HP_3PAR 7450c HP_3PAR 8200 HP_3PAR 8400 InServ F400 InServ T400 InServ T800 InServ V400	3.1.1 (MU1) 3.1.2 (MU3) 3.1.3 (MU1) 3.1.3 (MU2) 3.1.3 (MU3) 3.2.1 (MU3) 3.2.1 (MU5) 3.2.2 3.2.2 (MU2) 3.2.2 (MU4) 3.2.2 (MU6) 3.3.1 (MU1) 3.3.1 (MU2) 3.3.1 (MU5) 3.3.2 3.3.2 (MU1) 4.4.1 Release Type: Standard Support Release 4.5.11 Release Type: Extended Support Release 4.5.3 Release Type: Extended Support Release 4.5.7 Release Type: Extended Support Release 9.5.8 Release Type: Extended Support Release

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
2					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Storage Port	Implemented	SSH	
		Type	Gap	SSH	
	Volume Ref	Name	Implemented	SSH	
		Storage Id	Implemented	SSH	
	WWN Alias	Host Aliases	Implemented	SSH	
		Object Type	Implemented	SSH	
		Source	Implemented	SSH	
		WWN	Implemented	SSH	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Total	Implemented	SMI-S	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	SMI-S	
	Volume	Cache Hit Ratio Read	Implemented	SMI-S	
		Cache Hit Ratio Total	Implemented	SMI-S	
		Cache Hit Ratio Write	Implemented	SMI-S	
		Raw Capacity	Implemented	SMI-S	
		Total Capacity	Implemented	SMI-S	
		Used Capacity	Implemented	SMI-S	
		Capacity Used Ratio	Implemented	SMI-S	
		CapacityRatio Written	Implemented	SMI-S	
		IOps Read	Implemented	SMI-S	Number of read IOps on the disk
		IOPs Total	Implemented	SMI-S	
		IOPs Write	Implemented	SMI-S	
		Latency Read	Implemented	SMI-S	
		Latency Total	Implemented	SMI-S	
		Latency Write	Implemented	SMI-S	
		Partial Blocked Ratio	Implemented	SMI-S	
		Throughput Read	Implemented	SMI-S	
		Throughput Total	Implemented	SMI-S	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	SMI-S	
		Write Pending	Implemented	SMI-S	total write pending

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
3Par SMI-S	SMI-S	HTTP/HTTPS	5988/5989		true	true	true	true
3Par CLI	SSH	SSH	22		true	false	true	true

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Amazon AWS EC2

Models and versions supported by this data collector:

API versions
2014-10-01

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Manufacturer	Implemented	HTTPS	
		Name	Implemented	HTTPS	
		OID	Implemented	HTTPS	
	Info	Api Description	Implemented	HTTPS	
		Api Name	Implemented	HTTPS	
		Api Version	Implemented	HTTPS	
		DataSource Name	Implemented	HTTPS	Info
		Date	Implemented	HTTPS	
		Originator ID	Implemented	HTTPS	
		Originator Key	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
18					

		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Throughput Write	Implemented	HTTPS	
	vm	Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		Total CPU Utilization	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		diskIops.total	Implemented	HTTPS	
		Disk IOPs write	Implemented	HTTPS	
		Latency Read	Implemented	HTTPS	
		Latency Total	Implemented	HTTPS	
		Latency Write	Implemented	HTTPS	
		Disk Throughput Read	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	total disk throughput read
		Disk Throughput Write	Implemented	HTTPS	
		IP Throughput Read	Implemented	HTTPS	
		Throughput total	Implemented	HTTPS	IP throughput total
		ipThroughput.write	Implemented	HTTPS	
		Total Memory Utilization	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
EC2 API	HTTPS	HTTPS	443		true	true	true	true

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Amazon AWS S3

Models and versions supported by this data collector:

Models	Firmware versions
S3	2010-08-01

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					of all disks that are spare)
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
	Storage Pool	Include In Dwh Capacity	Implemented	HTTPS	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	HTTPS	
		Physical Disk Capacity (MB)	Implemented	HTTPS	used as raw capacity for storage pool
		Raid Group	Implemented	HTTPS	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	HTTPS	ratio to convert from usable capacity to raw capacity
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
performance	Internal Volume	Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		Total Objects	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
S3 API	HTTPS	HTTPS	443		true	true	true	true

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Microsoft Azure NetApp Files

Models and versions supported by this data collector:

API versions	Models
2019-06-01	Azure NetApp Files

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					of all disks that are spare)
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
	Storage Pool	Data Allocated Capacity	Gap	HTTPS	capacity allocated for data
		Data Used Capacity	Implemented	HTTPS	
		Include In Dwh Capacity	Implemented	HTTPS	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	HTTPS	
		Physical Disk Capacity (MB)	Implemented	HTTPS	used as raw capacity for storage pool
		Raid Group	Implemented	HTTPS	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	HTTPS	ratio to convert from usable capacity to raw capacity
		Status	Implemented	HTTPS	
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Total Used Capacity	Implemented	HTTPS	Total capacity in MB
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance	Internal Volume	Latency Total	Implemented		
		IOps Read	Implemented		Number of read IOps on the disk
		Latency Read	Implemented		
		IOPs other	Implemented		
		IOPs Write	Implemented		
		Throughput Read	Implemented		
		Throughput Write	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		IOPs Total	Implemented		
		Latency Write	Implemented		
		Total Capacity	Implemented		
		Used Capacity	Implemented		
		Snapshot Used Capacity Ratio	Implemented		Reported as a time series
		Capacity Used Ratio	Implemented		
		Total Data Capacity	Implemented		
		Data Used Capacity	Implemented		
		Snapshot Used Capacity	Implemented		
	StoragePool Disk	IOps Read	Implemented		Number of read IOps on the disk
		IOPs Total	Implemented		
		IOPs Write	Implemented		
		Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Azure Netapp Files REST API	HTTPS	HTTPS	443		true	true	true	true

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Brocade Fibre Channel Switches

Models and versions supported by this data collector:

Models	Firmware versions
178.0 183.0 Brocade 200E Brocade 300E Brocade 3900 Brocade 4024 Embedded Brocade 48000 Brocade 5000 Brocade 5100 Brocade 5300 Brocade 5480 Embedded Brocade 6505 Brocade 6510 Brocade 6520 Brocade 6548 Brocade 7800 Brocade 7840 Brocade DCX Brocade DCX-4S Backbone Brocade DCX8510-4 Brocade DCX8510-8 Brocade G610 Brocade G620 Brocade G630 Brocade G720 Brocade M5424 Embedded Brocade X6-4 Brocade X6-8 Brocade X7-4 Brocade X7-8	v5.3.2c v6.2.1b v6.2.2g v6.3.2 v6.4.1a v6.4.2 v6.4.2a v7.0.0 v7.0.1b v7.1.0c v7.3.0c v7.3.1d v7.4.1d v7.4.1f v7.4.2a v7.4.2c v7.4.2d v7.4.2g v7.4.2g_cvr_824494_01 v7.4.2h v7.4.2j1 v8.0.2a v8.0.2c v8.0.2d v8.1.2g v8.1.2j v8.1.2k v8.2.0 v8.2.0b v8.2.1c v8.2.1d v8.2.2a v8.2.2b v8.2.2c v8.2.2d v8.2.2d4 v8.2.3 v8.2.3a v8.2.3a1 v8.2.3b v8.2.3c v8.2.3c1 v9.0.0b v9.0.1a v9.0.1b4 v9.0.1c v9.0.1d v9.0.1e v9.0.1e1 v9.1.0b v9.1.1 v9.1.1_01 v9.1.1b

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		WWN	Implemented	SSH	
	Zone	Zone Name	Implemented	SSH	
	Zone Member	Type	Gap	SSH	
		WWN	Implemented	SSH	
	Zoning Capabilities	Active Configuration	Implemented	SSH	
		Configuration Name	Implemented	SSH	
		Default Zoning Behavior	Implemented	SSH	
		WWN	Implemented	SSH	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

		Average Frame Size	Implemented	SNMP	Average Frame size of traffic
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Traffic Rate	Implemented	SNMP	
		Total Traffic Rate	Implemented	SNMP	
		Traffic Rate	Implemented	SNMP	
		Traffic Utilization	Implemented	SNMP	
		Traffic Utilization	Implemented	SNMP	Total traffic utilization
		Traffic Utilization	Implemented	SNMP	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Brocade SNMP	SNMP	SNMPv1, SNMPv2, SNMPv3	161		true	true	true	true
Brocade SSH	SSH	SSH	22		false	false	true	true
Data source wizard configuration	Manual Entry				true	true	true	true

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Brocade Network Advisor HTTP

Models and versions supported by this data collector:

API versions	Models	Firmware versions
14.4.1 14.4.3 14.4.4 14.4.5	Brocade 5300 Brocade 6510 Brocade 6520 Brocade 6548 Brocade DCX 8510-8 Brocade G620 DS-6620B EMC Connectrix ED-DCX8510-8B	v7.2.1a v7.3.1a v7.4.1b v7.4.2d v8.2.3b v8.2.3c v9.0.1a v9.0.1b v9.0.1e1

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Switch Status	Implemented	HTTP/S	
		Type	Gap	HTTP/S	
		WWN	Implemented	HTTP/S	
	Unknown	Driver	Implemented	HTTP/S	
		Firmware	Implemented	HTTP/S	
		Manufacturer	Implemented	HTTP/S	
		Model	Implemented	HTTP/S	
		WWN	Implemented	HTTP/S	
	WWN Alias	Host Aliases	Implemented	HTTP/S	
		Object Type	Implemented	HTTP/S	
		Source	Implemented	HTTP/S	
		WWN	Implemented	HTTP/S	
	Zone	Zone Name	Implemented	HTTP/S	
	Zone Member	Type	Gap	HTTP/S	
		WWN	Implemented	HTTP/S	
	Zoning Capabilities	Active Configuration	Implemented	HTTP/S	
		Configuration Name	Implemented	HTTP/S	
		WWN	Implemented	HTTP/S	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance	port	bbCreditZero.total	Implemented	HTTP/S	
		BB Credit	Implemented	HTTP/S	
		bbCreditZeroMs	Implemented	HTTP/S	
		portErrors.class3Discard	Implemented	HTTP/S	
		portErrors.crc	Implemented	HTTP/S	
		Port Error	Implemented	HTTP/S	
		Port Error	Implemented	HTTP/S	Port errors due to short frame
		portErrors.linkFailure	Implemented	HTTP/S	Port Errors link failure
		Port Error	Implemented	HTTP/S	Port errors signal loss
		Port Error	Implemented	HTTP/S	Port error sync loss
		Port Error	Implemented	HTTP/S	port errors timeout discard
		Port Error	Implemented	HTTP/S	Total port errors
		Traffic Rate	Implemented	HTTP/S	
		Total Traffic Rate	Implemented	HTTP/S	
		Traffic Rate	Implemented	HTTP/S	
		Traffic Utilization	Implemented	HTTP/S	
		Traffic Utilization	Implemented	HTTP/S	Total traffic utilization
		Traffic Utilization	Implemented	HTTP/S	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Brocade Network Advisor REST API	HTTP/HTTPS	HTTP/HTTPS	80/443		true	true	true	true

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Brocade FOS REST

Models and versions supported by this data collector:

Models	Firmware versions
Brocade 6505 Brocade G720 Brocade X6-8	v8.2.3c v8.2.3c1 v9.0.1e1 v9.1.1b

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Switch Status	Implemented	HTTPS	
		Type	Gap	HTTPS	
		VSAN Enabled	Implemented	HTTPS	
		WWN	Implemented	HTTPS	
	Unknown	Driver	Implemented	HTTPS	
		Firmware	Implemented	HTTPS	
		Manufacturer	Implemented	HTTPS	
		Model	Implemented	HTTPS	
		WWN	Implemented	HTTPS	
	WWN Alias	Host Aliases	Implemented	HTTPS	
		Object Type	Implemented	HTTPS	
		Source	Implemented	HTTPS	
		WWN	Implemented	HTTPS	
	Zone	Zone Name	Implemented	HTTPS	
	Zone Member	Type	Gap	HTTPS	
		WWN	Implemented	HTTPS	
	Zoning Capabilities	Active Configuration	Implemented	HTTPS	
		Configuration Name	Implemented	HTTPS	
		Default Zoning Behavior	Implemented	HTTPS	
		WWN	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

		TX Frames	Implemented	HTTPS	traffic average frame size
		Traffic Rate	Implemented	HTTPS	
		Total Traffic Rate	Implemented	HTTPS	
		Traffic Rate	Implemented	HTTPS	
		Traffic Utilization	Implemented	HTTPS	
		Traffic Utilization	Implemented	HTTPS	Total traffic utilization
		Traffic Utilization	Implemented	HTTPS	

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Brocade FOS REST API	HTTPS		443		true	true	true	true

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Cisco MDS & Nexus Fabric Switches

Models and versions supported by this data collector:

Models	Firmware versions
8978-E04 CN1610 DS-C9124-2-K9 DS-C9124-K9 DS-C9132T-K9 DS-C9134-K9 DS-C9148-16P-K9 DS-C9148-32P-K9 DS-C9148-48P-K9 DS-C9148S-K9 DS-C9148T-K9 DS-C9222I-K9 DS-C9250I-K9 DS-C9396S-K9 DS-C9396T-K9 DS-C9506 DS-C9509 DS-C9513 DS-C9706 DS-C9710 DS-C9718 DS-HP-8GFC-K9 DS-HP-FC-K9 N5K-C5548UP N5K-C5596UP N5K-C56128P N5K-C5696Q UCS-FI-6248UP UCS-FI-6296UP UCS-FI-6332 UCS-FI-6332-16UP UCS-FI-6454	3.3(1c) 4.1(3a) 5.0(1a) 5.0(3)N2(3.11e) 5.0(3)N2(3.23o) 5.0(3)N2(4.01d) 5.0(3)N2(4.04e) 5.0(3)N2(4.13e) 5.0(3)N2(4.13i) 5.0(3)N2(4.21e) 5.0(3)N2(4.21j) 5.0(3)N2(4.21k) 5.0(3)N2(4.22c) 5.0(8) 5.2(2d) 5.2(3)N2(2.28g) 5.2(6a) 5.2(8) 5.2(8b) 5.2(8c) 5.2(8d) 5.2(8f) 5.2(8g) 5.2(8h) 5.2(8i) 6.2(1) 6.2(11) 6.2(11b) 6.2(11c) 6.2(11e) 6.2(13) 6.2(13a) 6.2(15) 6.2(17) 6.2(19) 6.2(21) 6.2(23) 6.2(25) 6.2(27) 6.2(29) 6.2(31) 6.2(33) 6.2(5) 6.2(5a) 6.2(7) 6.2(9) 6.2(9a) 6.2(9c) 7.3(0)D1(1) 7.3(0)DY(1) 7.3(1)DY(1) 7.3(1)N1(1) 7.3(13)N1(1) 7.3(6)N1(1)

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Switch Status	Implemented	SNMP	
		Type	Gap	SNMP	
		VSAN Enabled	Implemented	SNMP	
		WWN	Implemented	SNMP	
	Unknown	Driver	Implemented	SNMP	
		Firmware	Implemented	SNMP	
		Generated	Implemented	SNMP	
		Manufacturer	Implemented	SNMP	
		Model	Implemented	SNMP	
		Name	Implemented	SNMP	
		WWN	Implemented	SNMP	
	WWN Alias	Host Aliases	Implemented	SNMP	
		Object Type	Implemented	SNMP	
		Source	Implemented	SNMP	
		WWN	Implemented	SNMP	
	Zone	Zone Name	Implemented	SNMP	
		Zone Type	Implemented	SNMP	
	Zone Member	Type	Gap	SNMP	
		WWN	Implemented	SNMP	
	Zoning Capabilities	Active Configuration	Implemented	SNMP	
		Configuration Name	Implemented	SNMP	
		Default Zoning Behavior	Implemented	SNMP	
		Merge Control	Implemented	SNMP	
		WWN	Implemented	SNMP	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

		Traffic Rate	Implemented	SNMP	
		Total Traffic Rate	Implemented	SNMP	
		Traffic Rate	Implemented	SNMP	
		Traffic Utilization	Implemented	SNMP	
		Traffic Utilization	Implemented	SNMP	Total traffic utilization
		Traffic Utilization	Implemented	SNMP	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Cisco SNMP	SNMP	SNMPv1 (Inventory only), SNMPv2, SNMPv3	161		true	true	true	true

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Cohesity

Models and versions supported by this data collector:

Models	Firmware versions
C2500 C2505 C4000 Compute Node C4600 C5036 C5066 C6025 C6035 C6055 PXG1 UCS-C240M5H10	6.5.1f_release-20210913_13f6a4bf 6.5.1f_u1_release-20211027_9e4e40cb 6.6.0d_u6_release-20221204_c03629f0 6.8.1_release-20220807_6c9115ef 6.8.1_u1_release-20221022_6f58ed2a 6.8.1_u2_release-20230412_5ced2ed3 6.8.1_u3_release-20230509_1e641b74 7.0_u1_release-20230222_8995f044

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Type	Gap		
		Virtual	Implemented		Is this a storage virtualization device?
		Encrypted	Implemented		

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
					and write across all disks) in MB/s
	IOPs Total	Implemented			Latency Write
	Implemented			Utilization Total	Implemented
			StoragePool Disk	IOps Read	Implemented
		Number of read IOps on the disk		IOPs Write	Implemented
				Throughput Read	Implemented
				Throughput Write	Implemented
				Throughput Total	Implemented
		Average disk total rate (read and write across all disks) in MB/s		IOPs Total	Implemented

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Cohesity REST API	HTTPS	HTTPS	443		true	true	true	true

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EMC Celerra (SSH)

Models and versions supported by this data collector:

Models	Firmware versions
NS-480FC	5.5.38-1
NSX	6.0.65-2
VG8	7.1.76-4
VNX5200	7.1.79-8
VNX5300	7.1.83-2
VNX5400	8.1.21-266
VNX5600	8.1.21-303
VNX7600	8.1.9-155

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					of all disks that are spare)
		Virtual	Implemented	SSH	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
	Storage Pool	Data Allocated Capacity	Gap	SSH	capacity allocated for data
		Data Used Capacity	Implemented	SSH	
		Dedupe Enabled	Implemented	SSH	Is dedupe enabled on the storage pool
		Include In Dwh Capacity	Implemented	SSH	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	SSH	
		Raid Group	Implemented	SSH	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	SSH	ratio to convert from usable capacity to raw capacity
		Snapshot Allocated Capacity	Gap	SSH	Allocated capacity of snapshots in MB
		Snapshot Used Capacity	Implemented	SSH	
		Storage Pool Id	Implemented	SSH	
		Thin Provisioning Supported	Implemented	SSH	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	SSH	
		Total Used Capacity	Implemented	SSH	Total capacity in MB
		Type	Gap	SSH	
		Virtual	Implemented	SSH	Is this a storage virtualization device?

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Celerra CLI	SSH	SSH			true	false	true	true

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EMC CLARiiON (NaviCLI)

Models and versions supported by this data collector:

API versions	Models	Firmware versions
6.23 6.26 6.28 7.30 7.32 7.33	AX4-5F8 CX3-20f CX3-40f CX4-480 VNX5100 VNX5200 VNX5300 VNX5400 VNX5500 VNX5600 VNX5700 VNX5800 VNX7600 VNX8000	04.28.000.5.710 04.30.000.5.525 05.32.000.5.218 05.32.000.5.219 05.32.000.5.221 05.32.000.5.225 05.32.000.5.249 05.33.000.5.074 05.33.009.5.155 05.33.009.5.184 05.33.009.5.186 05.33.009.5.218 05.33.009.5.231 05.33.009.5.236 05.33.009.5.238 05.33.009.6.305 05.33.021.5.256 05.33.021.5.266 2.23.50.5.710 3.26.20.5.011 3.26.40.5.029

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Thin Provisioned	Implemented	CLI	
		Type	Gap	CLI	
		UUID	Implemented	CLI	
		Used Capacity	Implemented	CLI	
	Volume Map	LUN	Implemented	CLI	Name of the backend lun
		Protocol Controller	Implemented	CLI	
		Storage Port	Implemented	CLI	
		Type	Gap	CLI	
	Volume Mask	Initiator	Implemented	CLI	
		Protocol Controller	Implemented	CLI	
		Storage Port	Implemented	CLI	
		Type	Gap	CLI	
	Volume Member	Capacity	Implemented	CLI	Snapshot Used capacity in MB
		Name	Implemented	CLI	
		Rank	Implemented	CLI	
		Total Raw Capacity	Implemented	CLI	Total raw capacity (sum of all disks on the array)
		Redundancy	Implemented	CLI	Redundancy level
		Storage Pool Id	Implemented	CLI	
		Used Capacity	Implemented	CLI	
	WWN Alias	Host Aliases	Implemented	CLI	
		IP	Implemented	CLI	
		Object Type	Implemented	CLI	
		Source	Implemented	CLI	
		WWN	Implemented	CLI	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Write	Implemented	CLI	
		Read Utilization	Implemented	CLI	
		Utilization Total	Implemented	CLI	
		Utilization Write	Implemented	CLI	
	Volume	Cache Hit Ratio Read	Implemented	CLI	
		Cache Hit Ratio Total	Implemented	CLI	
		Cache Hit Ratio Write	Implemented	CLI	
		Raw Capacity	Implemented	CLI	
		Total Capacity	Implemented	CLI	
		Used Capacity	Implemented	CLI	
		Capacity Used Ratio	Implemented	CLI	
		IOps Read	Implemented	CLI	Number of read IOps on the disk
		IOPs Total	Implemented	CLI	
		IOPs Write	Implemented	CLI	
		Latency Read	Implemented	CLI	
		Latency Total	Implemented	CLI	
		Latency Write	Implemented	CLI	
		Partial Blocked Ratio	Implemented	CLI	
		Throughput Read	Implemented	CLI	
		Throughput Total	Implemented	CLI	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	CLI	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Navi CLI	CLI		6389,2162,2163,443(HTTPS)/80(HTTP)		true	true	true	false

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EMC Data Domain (SSH)

Models and versions supported by this data collector:

Models	Firmware versions
DD VE DD2200 DD2500 DD3300 DD4200 DD6300 DD6800 DD6900 DD7200 DD9300 DD9400 DD9500 DD9800 DD990 DD9900	6.1.2.051-633576 6.1.2.20-606786 6.1.2.50-632120 6.2.0.30-629757 6.2.0.35-635767 6.2.1.30-663869 6.2.1.40-671977 6.2.1.60-686365 7.10.0.0-1017741 7.10.1.0-1042928 7.2.0.30-663847 7.2.0.50-671975 7.2.0.60-682124 7.2.0.70-686759 7.2.0.90-692270 7.6.0.20-689174 7.6.0.30-690691 7.7.0.7-1007134 7.7.1.10-1011247 7.7.2.011-1011427 7.7.2.10-1011249 7.7.3.0-1011963 7.7.4.0-1017976 7.7.5.1-1040473 7.7.5.11-1046187 7.8.0.0-1008134

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					provisioning for the volume layer on top of it
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Total Used Capacity	Implemented	SSH	Total capacity in MB
		Type	Gap	SSH	
		Virtual	Implemented	SSH	Is this a storage virtualization device?

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Data Domain CLI	SSH	SSH	22		true	true	true	true

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EMC ECS

Models and versions supported by this data collector:

Models	Firmware versions
ECS	3.6.1.1 3.6.1.3 3.6.2.1 3.6.2.4 3.7.0.0 3.7.0.3 3.7.0.4 3.7.0.5 3.8.0.1 3.8.0.2

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Utilization Usable in MB			
		Node Capacity	Implemented	HTTPS	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Storage Pool	Include In Dwh Capacity	Implemented	HTTPS	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	HTTPS	
		Physical Disk Capacity (MB)	Implemented	HTTPS	used as raw capacity for storage pool
		Raid Group	Implemented	HTTPS	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	HTTPS	ratio to convert from usable capacity to raw capacity
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Total Used Capacity	Implemented	HTTPS	Total capacity in MB
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance	Internal Volume	Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		Total Objects	Implemented	HTTPS	
	Storage	IOPs Total	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
	Storage Node	IOPs Total	Implemented	HTTPS	
		Node Capacity Utilization Total	Implemented	HTTPS	
		Node Capacity Utilization Usable	Implemented	HTTPS	
		Node Capacity Utilization Used	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
	StoragePool Disk	Capacity Provisioned	Implemented	HTTPS	
		Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Over Commit Capacity Ratio	Implemented	HTTPS	Reported as a time series
		Capacity Used Ratio	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
EMC ECS REST API	HTTPS	HTTPS	443		true	true	true	true

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Dell EMC Isilon & PowerScale Rest

Models and versions supported by this data collector:

Models	Firmware versions
A200 A2000 A300 A3000 F200 F600 F800 F900 H400 H500 NL410 S210 X210 X400 X410	9.1.0.11 9.1.0.6 9.2.1.10 9.2.1.11 9.2.1.12 9.2.1.16 9.2.1.19 9.2.1.21 9.2.1.6 9.2.1.7 9.2.1.9 9.4.0.11 9.4.0.12 9.4.0.13 9.4.0.14 9.4.0.5 9.4.0.7 9.5.0.3 v8.0.0.4 v8.0.0.6 v8.0.0.7 v8.1.2.0 v8.2.2.0

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Capacity			snapshots in MB
		Snapshot Used Capacity	Implemented	HTTPS	
Product	**Category**	**Feature/Attribut**	**Status**	**Protocol Used**	**Additional Information**
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Total Used Capacity	Implemented	HTTPS	Total capacity in MB
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
					and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
		Utilization Total	Implemented	HTTPS	
	StoragePool Disk	Capacity Provisioned	Implemented	HTTPS	
		Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Over Commit Capacity Ratio	Implemented	HTTPS	Reported as a time series
		Capacity Used Ratio	Implemented	HTTPS	
		Total Data Capacity	Implemented	HTTPS	
		Data Used Capacity	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Other Total Capacity	Implemented	HTTPS	
		Other Used Capacity	Implemented	HTTPS	
		Snapshot Reserved Capacity	Implemented	HTTPS	
		Snapshot Used Capacity	Implemented	HTTPS	
		Snapshot Used Capacity Ratio	Implemented	HTTPS	Reported as a time series
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
EMC Isilon & PowerScale REST API	HTTPS		443		true	true	true	true

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Dell EMC Isilon / PowerScale (CLI)

Models and versions supported by this data collector:

Models	Firmware versions
A200 A2000 A300 F200 F800 F900 H400 H500 H600 H700 NL400 NL410 S210 X200 X210 X400 X410	9.1.0.10 9.1.0.12 9.1.0.16 9.1.0.18 9.1.0.19 9.1.0.7 9.2.1.11 9.2.1.13 9.2.1.15 9.2.1.22 9.2.1.7 9.2.1.9 9.3.0.3 9.4.0.0 9.4.0.10 9.4.0.12 9.4.0.13 9.4.0.14 9.4.0.6 9.4.0.7 v7.1.1.8 v7.2.0.5 v7.2.1.3 v7.2.1.6 v8.0.0.4 v8.0.0.6 v8.0.0.7 v8.0.1.1 v8.1.2.0 v8.2.2.0

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					device state
		UUID	Implemented	SSH	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Management IP Addresses	Implemented	SSH	
	Storage Pool	Data Allocated Capacity	Gap	SSH	capacity allocated for data
		Data Used Capacity	Implemented	SSH	
		Include In Dwh Capacity	Implemented	SSH	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	SSH	
		Physical Disk Capacity (MB)	Implemented	SSH	used as raw capacity for storage pool
		Raid Group	Implemented	SSH	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	SSH	ratio to convert from usable capacity to raw capacity
		Snapshot Allocated Capacity	Gap	SSH	Allocated capacity of snapshots in MB
		Snapshot Used Capacity	Implemented	SSH	
		Storage Pool Id	Implemented	SSH	
		Thin Provisioning Supported	Implemented	SSH	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	SSH	
		Total Used Capacity	Implemented	SSH	Total capacity in MB
		Type	Gap	SSH	
		Virtual	Implemented	SSH	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
124					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					and write across all disks) in MB/s
		Throughput Write	Implemented	SSH	
		Utilization Total	Implemented	SSH	
	StoragePool Disk	Capacity Provisioned	Implemented	SSH	
		Raw Capacity	Implemented	SSH	
		Total Capacity	Implemented	SSH	
		Used Capacity	Implemented	SSH	
		Over Commit Capacity Ratio	Implemented	SSH	Reported as a time series
		Capacity Used Ratio	Implemented	SSH	
		Total Data Capacity	Implemented	SSH	
		Data Used Capacity	Implemented	SSH	
		Snapshot Reserved Capacity	Implemented	SSH	
		Snapshot Used Capacity	Implemented	SSH	
		Snapshot Used Capacity Ratio	Implemented	SSH	Reported as a time series

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Isilon SSH	SSH	SSH	22		true	false	true	true

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EMC PowerStore REST

Models and versions supported by this data collector:

Models	Firmware versions
PowerStore 1000T PowerStore 1200T PowerStore 3000T PowerStore 3200T PowerStore 5000T PowerStore 5000X PowerStore 9000T PowerStore 9200T	2.0.1.3 2.1.1.0 2.1.1.1 3.0.0.1 3.2.0.0 3.2.0.1 3.2.1.0

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Type	Gap		
		Virtual	Implemented		Is this a storage virtualization device?
	Volume	Capacity	Implemented		Snapshot Used capacity in MB
		Junction Path	Implemented		
		Name	Implemented		
		Total Raw Capacity	Implemented		Total raw capacity (sum of all disks on the array)
		Storage Pool Id	Implemented		
		Thin Provisioned	Implemented		
		Type	Gap		
		UUID	Implemented		
		Used Capacity	Implemented		
		QoS - Policy	Implemented		
	Volume Map	LUN	Implemented		Name of the backend lun
		Masking Required	Implemented		
		Protocol Controller	Implemented		
		Storage Port	Implemented		
		Type	Gap		
	Volume Mask	Initiator	Implemented		
		Protocol Controller	Implemented		
		Type	Gap		
	WWN Alias	Host Aliases	Implemented		
		Object Type	Implemented		
		Source	Implemented		
		WWN	Implemented		

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
136					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

	Implemented				Throughput Total
	Implemented		Average disk total rate (read and write across all disks) in MB/s		Throughput Write

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
	Implemented			Volume	Raw Capacity
	Implemented				Total Capacity
	Implemented				Used Capacity
	Implemented				Capacity Used Ratio
	Implemented				IOps Read
	Implemented		Number of read IOps on the disk		IOPs Total
	Implemented				IOPs Write
	Implemented				Latency Read
	Implemented				Latency Total
	Implemented				Latency Write
	Implemented				Throughput Read
	Implemented				Throughput Total
	Implemented		Average disk total rate (read and write across all disks) in MB/s		Throughput Write

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
EMC PowerStore REST API	HTTPS	HTTPS	443		true	true	true	true

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EMC RecoverPoint (HTTP)

Models and versions supported by this data collector:

Models	Firmware versions
RecoverPoint	5.1.P1(c.175) 5.1.SP4.P1(h.89) 5.1.SP4.P2(h.101) 5.1.SP4.P3(h.109) 5.1.SP4.P4(h.97)

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
140					

					of all disks that are spare)
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Storage Node	Memory Size	Gap	HTTPS	device memory in MB
		Model	Implemented	HTTPS	
		Name	Implemented	HTTPS	
		Processors Count	Implemented	HTTPS	device CPU
		Serial Number	Implemented	HTTPS	
		State	Implemented	HTTPS	free text describing the device state
		UUID	Implemented	HTTPS	
		Version	Implemented	HTTPS	software version
	Storage Synchronization	Mode	Implemented	HTTPS	
		Mode Enum	Implemented	HTTPS	
		Source Storage	Implemented	HTTPS	
		Source Volume	Implemented	HTTPS	
		State	Implemented	HTTPS	free text describing the device state
		State Enum	Implemented	HTTPS	
		Target Storage	Implemented	HTTPS	
		Target Volume	Implemented	HTTPS	
		Technology	Implemented	HTTPS	technology which causes storage efficiency changed

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
RecoverPoint REST API	HTTPS	HTTPS	443		true	true	true	true

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EMC ScaleIO & PowerFlex REST

Models and versions supported by this data collector:

Models	Firmware versions
ScaleIO	R2_6.11000.113 R2_6.11000.115 R3_0.1400.101 R3_5.1200.104 R3_6.500.113 R3_6.700.103

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Volume	Capacity	Implemented	HTTPS	Snapshot Used capacity in MB
		Junction Path	Implemented	HTTPS	
		Name	Implemented	HTTPS	
		Total Raw Capacity	Implemented	HTTPS	Total raw capacity (sum of all disks on the array)
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioned	Implemented	HTTPS	
		UUID	Implemented	HTTPS	
		Host IPs	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
	Storage Node	IOps Read	Implemented		Number of read IOps on the disk
		IOPs Total	Implemented		
		IOPs Write	Implemented		
		Latency Read	Implemented		
		Latency Total	Implemented		
		Latency Write	Implemented		
		Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
	StoragePool Disk	IOps Read	Implemented		Number of read IOps on the disk
		IOPs Write	Implemented		
		Throughput Read	Implemented		
		Throughput Write	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		IOPs Total	Implemented		
	Volume	Raw Capacity	Implemented		
		Total Capacity	Implemented		
		IOps Read	Implemented		Number of read IOps on the disk
		IOPs Total	Implemented		
		IOPs Write	Implemented		
		Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
EMC ScaleIO & PowerFlex REST API	HTTPS	HTTPS	443		true	true	true	true

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EMC Symmetrix CLI

Models and versions supported by this data collector:

API versions	Models	Firmware versions
V10.0.0.0 V10.0.1.0 V7.6.2.67 V8.3.0.22 V8.3.0.6 V8.4.0.7 V8.4.0.9 V9.1.0.18 V9.1.0.5 V9.1.0.6 V9.2.0.0 V9.2.1.0 V9.2.1.1 V9.2.1.2 V9.2.2.0 V9.2.3.0 V9.2.3.1 V9.2.3.4 V9.2.3.5 V9.2.3.6 V9.2.4.1 V9.2.4.2	DMX3-24 DMX4-24 PMax2000 PowerMax_2000 PowerMax_8000 VMAX-1 VMAX100K VMAX10K VMAX200K VMAX250F VMAX400K VMAX40K VMAX450F VMAX850F VMAX950F	5773.198.142(168D0000) build 5 5876.272.177(16F40000) build 39 5876.286.194(16F40000) build 115 5876.309.196(16F40000) build 162 5977.1131.1131(17590000) build 551 5977.1151.1151(17590000) build 45 5977.1151.1151(17590000) build 59 5977.1151.1151(17590000) build 60 5977.1151.1151(17590000) build 9 5978.479.479(175A0000) build 195 5978.711.711(175A0000) build 113 5978.711.711(175A0000) build 139 5978.711.711(175A0000) build 149 5978.711.711(175A0000) build 194 5978.711.711(175A0000) build 196 5978.711.711(175A0000) build 220 5978.711.711(175A0000) build 239 5978.711.711(175A0000) build 252 5978.711.711(175A0000) build 267 5978.711.711(175A0000) build 278 5978.711.711(175A0000) build 287 5978.711.711(175A0000) build 335 5978.711.711(175A0000) build 365 5978.711.711(175A0000) build 366 5978.711.711(175A0000) build 388 5978.711.711(175A0000) build 416 5978.711.711(175A0000) build 436 5978.711.711(175A0000) build 438 5978.711.711(175A0000) build 448 5978.711.711(175A0000) build 461 5978.711.711(175A0000) build 480 5978.711.711(175A0000) build 484 5978.711.711(175A0000) build 502 5978.711.711(175A0000) build 529 5978.711.711(175A0000) build 8

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Storage Pool Id	Implemented		
		UUID	Implemented		
		Used Capacity	Implemented		
	Volume Ref	Name	Implemented		
		Storage Ip	Implemented		
	WWN Alias	Host Aliases	Implemented		
		Object Type	Implemented		
		Source	Implemented		
		WWN	Implemented		

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
160					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
	Volume	Cache Hit Ratio Read	Implemented		
		Cache Hit Ratio Total	Implemented		
		Cache Hit Ratio Write	Implemented		
		Raw Capacity	Implemented		
		Total Capacity	Implemented		
		Used Capacity	Implemented		
		Written Capacity	Implemented		
		Capacity Used Ratio	Implemented		
		CapacityRatio Written	Implemented		
		IOps Read	Implemented		Number of read IOps on the disk
		IOPs Total	Implemented		
		IOPs Write	Implemented		
		Latency Read	Implemented		
		Latency Total	Implemented		
		Latency Write	Implemented		
		Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
		Write Pending	Implemented		total write pending

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
symcli	CLI		2707		true	true	true	true

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Symmetrix SMI-S	SMI-S	HTTP/HTTPS	5988/5989		true	false	false	true

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Dell Unisphere REST

Models and versions supported by this data collector:

API versions	Models	Firmware versions
V10.0.0.5 V10.0.1.3 V9.2.1.6 V9.2.3.20 V9.2.3.22 V9.2.3.4 V9.2.4.1	PowerMax_2000 PowerMax_2500 PowerMax_8000 VMAX250F VMAX950F	5978.479.479 build 350 5978.711.711 build 252 5978.711.711 build 278 build 278 5978.711.711 build 287 5978.711.711 build 329 build 329 5978.711.711 build 365 5978.711.711 build 365 build 365 5978.711.711 build 376 5978.711.711 build 388 build 388 5978.711.711 build 416 5978.711.711 build 435 5978.711.711 build 448 5978.711.711 build 461 build 461 5978.711.711 build 481 build 481 5978.711.711 build 484 5978.711.711 build 484 build 484 5978.711.711 build 502 6079.125.0 build 53 build 53 6079.175.0 build 0 build 0

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
166					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
		storage Groups	Implemented	HTTPS	
	Volume Map	LUN	Implemented	HTTPS	Name of the backend lun
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	WWN Alias	Host Aliases	Implemented	HTTPS	
		Object Type	Implemented	HTTPS	
		Source	Implemented	HTTPS	
		WWN	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

		Total Compression Savings	Implemented	HTTPS	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Compression Savings Space		HTTPS	Total Data Capacity	
	Implemented	HTTPS		Data Used Capacity	Implemented
	HTTPS		IOps Read	Implemented	HTTPS
	Number of read IOps on the disk	IOPs Total	Implemented	HTTPS	
	IOPs Write	Implemented	HTTPS		Snapshot Reserved Capacity
	Implemented	HTTPS		Snapshot Used Capacity Ratio	Implemented
	HTTPS	Reported as a time series	Throughput Read	Implemented	HTTPS
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
	Throughput Write	Implemented	HTTPS		Utilization Total
	Implemented	HTTPS		Volume	Raw Capacity
	Implemented	HTTPS			Total Capacity
	Implemented	HTTPS			Used Capacity
	Implemented	HTTPS			Capacity Used Ratio
	Implemented	HTTPS			CapacityRatio Written
	Implemented	HTTPS			IOps Read
	Implemented	HTTPS	Number of read IOps on the disk		IOPs Total
	Implemented	HTTPS			IOPs Write
	Implemented	HTTPS			Latency Read
	Implemented	HTTPS			Latency Total
	Implemented	HTTPS			Latency Write
	Implemented	HTTPS			Throughput Read
	Implemented	HTTPS			Throughput Total
	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s		Throughput Write

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Dell Unisphere API	HTTPS	HTTPS	443		true	true	true	true

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EMC VNX (SSH)

Models and versions supported by this data collector:

Models	Firmware versions
VNX5300 VNX5400 VNX5700 VNX5800	05.33.009.5.231 7.1.76-4 7.1.80-3 8.1.9-232

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
174					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Storage Port	Implemented	SSH	
		Protocol Controller	Implemented	SSH	
		Type	Gap	SSH	
	Volume Mask	Storage Port	Implemented	SSH	
		Initiator	Implemented	SSH	
		Protocol Controller	Implemented	SSH	
		Type	Gap	SSH	
	WWN Alias	Source	Implemented	SSH	
		Host Aliases	Implemented	SSH	
		WWN	Implemented	SSH	
		Object Type	Implemented	SSH	
		IP	Implemented	SSH	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

		Throughput Total	Implemented	SSH	Average disk total rate (read and write across all disks) in MB/s
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Throughput Write	Implemented	SSH	
	Storage Node	IOps Read	Implemented	SSH	Number of read IOps on the disk
		IOPs Total	Implemented	SSH	
		IOPs Write	Implemented	SSH	
		Utilization Total	Implemented	SSH	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
VNX SSH and CLI	SSH	SSH	22		true	false	true	true

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EMC VNXe & Unity Unisphere (CLI)

Models and versions supported by this data collector:

Models	Firmware versions
Unity 300 Unity 300F Unity 350F Unity 380 Unity 380F Unity 400 Unity 400F Unity 450F Unity 480F Unity 500 Unity 550F Unity 600 Unity 600F Unity 650F Unity 680F Unity 880 VNXe3200	3.1.17.10223906 3.1.17.10229825 4.1.2.9257522 4.2.1.9535982 4.2.3.9670635 4.5.1.0.5.001 5.0.2.0.5.009 5.0.6.0.5.008 5.0.8.0.5.007 5.1.2.0.5.007 5.1.3.0.5.003 5.2.1.0.5.013 5.2.2.0.5.004 5.2.2.0.6.201 5.3.0.0.5.120

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
188					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
196					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

	Implemented	HTTPS			Throughput Total
	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s		Throughput Write

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
	Implemented	HTTPS		Volume	Raw Capacity
	Implemented	HTTPS			Total Capacity
	Implemented	HTTPS			Used Capacity
	Implemented	HTTPS			Capacity Used Ratio
	Implemented	HTTPS			IOps Read
	Implemented	HTTPS	Number of read IOps on the disk		IOPs Total
	Implemented	HTTPS			IOPs Write
	Implemented	HTTPS			Latency Total
	Implemented	HTTPS			Throughput Read
	Implemented	HTTPS			Throughput Total
	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s		Throughput Write

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
VNXe & Unisphere CLI	HTTPS	HTTPS	443		true	true	true	true

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EMC VPLEX

Models and versions supported by this data collector:

Models	Firmware versions
VPLEX	5.4.1.00.00.07 5.4.1.01.00.05 6.2.0.03.00.02 6.2.0.04.00.07 6.2.0.05.00.11 6.2.0.07.00.02

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
200					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Protocol Controller	Implemented	HTTP/S	
		Storage Port	Implemented	HTTP/S	
		Type	Gap	HTTP/S	
	Volume Mask	Initiator	Implemented	HTTP/S	
		Protocol Controller	Implemented	HTTP/S	
		Storage Port	Implemented	HTTP/S	
		Type	Gap	HTTP/S	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
204					

					and write across all disks) in MB/s
		Throughput Write	Implemented	SSH	
		Utilization Total	Implemented	SSH	
	StoragePool Disk	Capacity Provisioned	Implemented	SSH	
		Total Capacity	Implemented	SSH	
		Used Capacity	Implemented	SSH	
		Over Commit Capacity Ratio	Implemented	SSH	Reported as a time series
		Capacity Used Ratio	Implemented	SSH	
		Other Total Capacity	Implemented	SSH	
		Other Used Capacity	Implemented	SSH	
	Volume	Raw Capacity	Implemented	SSH	
		Total Capacity	Implemented	SSH	
		IOPs Total	Implemented	SSH	
		Latency Read	Implemented	SSH	
		Latency Total	Implemented	SSH	
		Latency Write	Implemented	SSH	
		Throughput Read	Implemented	SSH	
		Throughput Total	Implemented	SSH	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	SSH	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
EMC VPLEX CLI	SSH	SSH	22		true	true	true	true
EMC VPLEX API	HTTP/HTTPS	HTTP/HTTPS	80/443		true	true	true	true

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EMC XtremIO (HTTP)

Models and versions supported by this data collector:

API versions	Models	Firmware versions
6.2.1 6.2.2 6.3.1 6.3.2 6.3.3 6.4.0	1 Bricks & 125TB 1 Bricks & 24TB 1 Bricks & 26TB 1 Bricks & 31TB 1 Bricks & 62TB 1 Bricks & 8TB 1X10TB 1X20TB 1X40TB 2 Bricks & 52TB 2 Bricks & 62TB 2 Bricks & 76TB 2 Bricks & 83TB 2X10TB 2X20TB 2X40TB 3 Bricks & 251TB 3 Bricks & 283TB 4 Bricks & 125TB 4 Bricks & 503TB 4 Bricks & 628TB 4 Bricks & 754TB 4X20TB 4X40TB 6X20TB	4.0.27-1 4.0.31-11 6.1.0-99_X2 6.3.3-8_X2 6.4.0-22_X2 6.4.0-36_hotfix_2_X2

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Target Volume	Implemented	HTTPS	
		Technology	Implemented	HTTPS	technology which causes storage efficiency changes
	Volume	Capacity	Implemented	HTTPS	Snapshot Used capacity in MB
		Disk Size	Implemented	HTTPS	comma seperated list of disk sizes (GB)
		Disk Speed	Implemented	HTTPS	comma seperated list of disk speeds (rpm)
		Disk Type	Not Available	HTTPS	
		Name	Implemented	HTTPS	
		Total Raw Capacity	Implemented	HTTPS	Total raw capacity (sum of all disks on the array)
		Redundancy	Implemented	HTTPS	Redundancy level
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioned	Implemented	HTTPS	
		Type	Gap	HTTPS	
		UUID	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
	Volume Map	LUN	Implemented	HTTPS	Name of the backend lun
		Protocol Controller	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Type	Gap	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
	StoragePool Disk	Capacity Provisioned	Implemented	HTTPS	
		Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Over Commit Capacity Ratio	Implemented	HTTPS	Reported as a time series
		Capacity Used Ratio	Implemented	HTTPS	
		Total Data Capacity	Implemented	HTTPS	
		Data Used Capacity	Implemented	HTTPS	
	Volume	Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Latency Read	Implemented	HTTPS	
		Latency Total	Implemented	HTTPS	
		Latency Write	Implemented	HTTPS	
		Partial Blocked Ratio	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
EMC XtremIO REST API	HTTPS	HTTPS	443		true	true	true	true

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NetApp E-Series

Models and versions supported by this data collector:

Models	Firmware versions
2600 2660 2680 2702 2704 2800B 2804 2806 3000 5480 5486 5488 5504 5564 5600 5700 5700B 6000	08.40.60.01 8.10.14.0 8.20.11.0 8.20.27.0 8.20.30.0 8.20.5.0 8.20.8.0 8.25.14.0 8.25.6.0 8.30.1.0 8.40.0.1 8.40.0.3 8.40.20.0 8.40.30.3 8.40.40.0 8.40.50.0 8.40.60.1 8.40.60.2 8.40.60.3 8.42.20.0 8.50.0.3 8.50.0.4 8.51.0.0 8.52.0.0 8.52.0.1 8.53.0.1 8.53.0.4 8.62.0.0 8.62.0.2 8.63.0.2 8.70.0.3 8.71.2.0 8.71.3.0 8.72.0.0 8.72.1.0 8.72.2.0 8.73.0.0 8.74.0.0 8.74.1.0 8.74.2.0 8.74.3.0 8.75.0.0

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
218					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Supported			internal volume supports thin provisioning for the array layer on top of it
		Total Allocated Capacity	Implemented	RMI	
		Total Used Capacity	Implemented	RMI	Total capacity in MB
		Type	Gap	RMI	
		Virtual	Implemented	RMI	Is this a storage virtualization device?
	Volume	Capacity	Implemented	RMI	Snapshot Used capacity in MB
		Disk Type	Not Available	RMI	
		Name	Implemented	RMI	
		Total Raw Capacity	Implemented	RMI	Total raw capacity (sum of all disks on the array)
		Redundancy	Implemented	RMI	Redundancy level
		Storage Pool Id	Implemented	RMI	
		Thin Provisioned	Implemented	RMI	
		Type	Gap	RMI	
		UUID	Implemented	RMI	
		Used Capacity	Implemented	RMI	
		Virtual	Implemented	RMI	Is this a storage virtualization device?
		Written Capacity	Implemented	RMI	Total capacity written to this volume by a Host in MB
	Volume Map	LUN	Implemented	RMI	Name of the backend lun
		Storage Port	Implemented	RMI	
		Type	Gap	RMI	
	Volume Mask	Initiator	Implemented	RMI	
		Storage Port	Implemented	RMI	
		Type	Gap	RMI	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Read Utilization	Implemented	RMI	
		Utilization Total	Implemented	RMI	
		Utilization Write	Implemented	RMI	
	Volume	Cache Hit Ratio Read	Implemented	RMI	
		Cache Hit Ratio Total	Implemented	RMI	
		Cache Hit Ratio Write	Implemented	RMI	
		Raw Capacity	Implemented	RMI	
		Total Capacity	Implemented	RMI	
		Used Capacity	Implemented	RMI	
		Written Capacity	Implemented	RMI	
		Capacity Used Ratio	Implemented	RMI	
		CapacityRatio Written	Implemented	RMI	
		IOps Read	Implemented	RMI	Number of read IOps on the disk
		IOPs Total	Implemented	RMI	
		IOPs Write	Implemented	RMI	
		Latency Read	Implemented	RMI	
		Latency Total	Implemented	RMI	
		Latency Write	Implemented	RMI	
		Throughput Read	Implemented	RMI	
		Throughput Total	Implemented	RMI	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	RMI	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
SANtricity API	RMI	TCP			true	true	false	false

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Google Cloud Compute

Models and versions supported by this data collector:

API versions
v1

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Manufacturer	Implemented	HTTPS	
		Name	Implemented	HTTPS	
		OID	Implemented	HTTPS	
	Info	Api Description	Implemented	HTTPS	
		Api Name	Implemented	HTTPS	
		Api Version	Implemented	HTTPS	
		DataSource Name	Implemented	HTTPS	Info
		Date	Implemented	HTTPS	
		Originator ID	Implemented	HTTPS	
		Originator Key	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
	vm	Total Capacity	Implemented	HTTPS	
		Total CPU Utilization	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		diskIops.total	Implemented	HTTPS	
		Disk IOPs write	Implemented	HTTPS	
		Latency Total	Implemented	HTTPS	
		Disk Throughput Read	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	total disk throughput read
		Disk Throughput Write	Implemented	HTTPS	
		IP Throughput Read	Implemented	HTTPS	
		Throughput total	Implemented	HTTPS	IP throughput total
		ipThroughput.write	Implemented	HTTPS	
		Total Memory Utilization	Implemented	HTTPS	
		swapRate.inRate	Implemented	HTTPS	
		Swap Rate	Implemented	HTTPS	
		Total Swap Rate	Implemented	HTTPS	
		Schedule wait time	Implemented	HTTPS	Waiting to be scheduled time in percent

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Google Compute Platform REST API	HTTPS		443		true	true	true	true

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HDS HCP (HTTPS)

Models and versions supported by this data collector:

Models	Firmware versions
Hitachi Content Platform	9.3.7.2 9.5.0.121

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
					virtualization device?
	Storage Node	Name	Implemented	HTTPS	
		UUID	Implemented	HTTPS	
	Storage Pool	Include In Dwh Capacity	Implemented	HTTPS	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	HTTPS	
		Physical Disk Capacity (MB)	Implemented	HTTPS	used as raw capacity for storage pool
		Raid Group	Implemented	HTTPS	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	HTTPS	ratio to convert from usable capacity to raw capacity
		Soft Limit (MB)	Implemented	HTTPS	logical volume size that is defined during volume creation or resizing operations
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Total Used Capacity	Implemented	HTTPS	Total capacity in MB
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance	Internal Volume	Total Objects	Implemented		
	Storage	Failed Raw Capacity	Implemented		
		Raw Capacity	Implemented		
		Spare Raw Capacity	Implemented		Raw capapcity of spare disks (sum of all disks that are spare)
		StoragePools Capacity	Implemented		
		Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
	Storage Node	Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
		Utilization Total	Implemented		
	StoragePool Disk	Total Capacity	Implemented		
		Capacity Used Ratio	Implemented		
		Capacity Provisioned	Implemented		
		Used Capacity	Implemented		
		Raw Capacity	Implemented		
		Capacity Soft Limit	Implemented		
		Over Commit Capacity Ratio	Implemented		Reported as a time series

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
HDS HCP REST API	HTTPS	HTTPS	9090		true	true	true	true

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HiCommand Device Manager

Models and versions supported by this data collector:

API versions	Models	Firmware versions
7.6.1 8.7.7 8.8.1 8.8.3 8.8.5	DF850MH DF850S HM800 HM850 P9500 RAID700 RAID800 VSP5000 XP24000 XP7	0983/A-H 0988/H-S DKC:60-08-22 DKC:60-08-65 DKC:70-06-46 DKC:70-06-67-00/00 DKC:80-06-80 DKC:80-06-82-00/00 DKC:80-06-86-00/00 DKC:80-06-87 DKC:80-06-88-00/00 DKC:80-06-91 DKC:80-06-91-00/00 DKC:80-06-93-00/00 DKC:83-05-45-40/00 DKC:83-05-45-60/00 DKC:83-05-46-60/00 DKC:83-05-47-60/00 DKC:83-05-48-40/00 DKC:83-05-48-60/00 DKC:88-08-08-60/00 DKC:88-08-09-60/00 DKC:90-08-81-00/00 DKC:90-08-83-00/01 SVP:60-08-21/00 SVP:60-08-54/00 SVP:70-06-32/00 SVP:70-06-51/00 SVP:80-06-76/02 SVP:80-06-78/00 SVP:80-06-81/00 SVP:80-06-82/00 SVP:80-06-83/00 SVP:80-06-86/00 SVP:80-06-88/00 SVP:83-05-49-40/00 SVP:83-05-49-60/00 SVP:83-05-50-60/00 SVP:83-05-51-60/00 SVP:83-05-52-40/00 SVP:83-05-52-60/00 SVP:88-08-10-60/00 SVP:88-08-11-60/00 SVP:90-08-81/00 SVP:90-08-83/00

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Used Capacity	Implemented	HDS XML API	
		Virtual	Implemented	HDS XML API	Is this a storage virtualization device?
	Volume Map	LUN	Implemented	HDS XML API	Name of the backend lun
		Masking Required	Implemented	HDS XML API	
		Protocol Controller	Implemented	HDS XML API	
		Storage Port	Implemented	HDS XML API	
	Volume Mask	Initiator	Implemented	HDS XML API	
		Protocol Controller	Implemented	HDS XML API	
		Storage Port	Implemented	HDS XML API	
	Volume Member	Name	Implemented	HDS XML API	
		Storage Pool Id	Implemented	HDS XML API	
		Rank	Implemented	HDS XML API	
		Cylinders	Implemented	HDS XML API	
		Capacity	Implemented	HDS XML API	Snapshot Used capacity in MB
		Total Raw Capacity	Implemented	HDS XML API	Total raw capacity (sum of all disks on the array)
		Used Capacity	Implemented	HDS XML API	
	WWN Alias	Host Aliases	Implemented	HDS XML API	
		Object Type	Implemented	HDS XML API	
		Source	Implemented	HDS XML API	
		WWN	Implemented	HDS XML API	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
248					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

	Storage Node	Throughput Total	Implemented	Export/CLI	Average disk total rate (read and write across all disks) in MB/s
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		IOPs Total	Implemented	Export/CLI	
	StoragePool Disk	Total Capacity	Implemented	Export/CLI	
		Capacity Used Ratio	Implemented	Export/CLI	
		Capacity Provisioned	Implemented	Export/CLI	
		Used Capacity	Implemented	Export/CLI	
		Raw Capacity	Implemented	Export/CLI	
		Capacity Soft Limit	Implemented	Export/CLI	
		Over Commit Capacity Ratio	Implemented	Export/CLI	Reported as a time series
	Volume	Latency Total	Implemented	Export/CLI	
		IOps Read	Implemented	Export/CLI	Number of read IOps on the disk
		Latency Read	Implemented	Export/CLI	
		Cache Hit Ratio Read	Implemented	Export/CLI	
		IOPs Write	Implemented	Export/CLI	
		Cache Hit Ratio Total	Implemented	Export/CLI	
		Cache Hit Ratio Write	Implemented	Export/CLI	
		Throughput Read	Implemented	Export/CLI	
		Throughput Write	Implemented	Export/CLI	
		Throughput Total	Implemented	Export/CLI	Average disk total rate (read and write across all disks) in MB/s
		IOPs Total	Implemented	Export/CLI	
		Latency Write	Implemented	Export/CLI	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Export utility (USPV) / SNM CLI (AMS)	Export/CLI				false	false	false	false
HiCommand Device Manager XML API	HDS XML API	HTTP/HTTPS	2001		true	true	true	true

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Hitachi Ops Center

Models and versions supported by this data collector:

Models	Firmware versions
VSP 5100 VSP 5500 VSP F1500 VSP F600 VSP G800	80-06-92-00/00:01-65-03/05 83-05-46-60/00:01-65-03/05 83-05-47-40/00:01-65-03/05 83-05-48-40/00:01-65-03/05 90-08-81-00/00:01-65-03/05 90-08-82-00/00:01-65-03/05

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
252					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Type	Gap		
		Used Capacity	Implemented		
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Enabled	Implemented		
	Volume Map	LUN	Implemented		Name of the backend lun
		Masking Required	Implemented		
		Protocol Controller	Implemented		
		Storage Port	Implemented		
		Type	Gap		
	Volume Mask	Initiator	Implemented		
		Protocol Controller	Implemented		
		Storage Port	Implemented		
		Type	Gap		

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

	Storage Node	Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		IOPs Total	Implemented		
	StoragePool Disk	Total Capacity	Implemented		
		Capacity Used Ratio	Implemented		
		Capacity Provisioned	Implemented		
		Used Capacity	Implemented		
		Raw Capacity	Implemented		
		Capacity Soft Limit	Implemented		
		Over Commit Capacity Ratio	Implemented		Reported as a time series

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Hitachi Ops Center REST API	HTTPS	HTTPS	443		true	true	true	true

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HDS HNAS (CLI)

Models and versions supported by this data collector:

Models	Firmware versions
G600 G800 HNAS 4080 HNAS 4100 N800	13.9.6918.05 14.5.7413.01 14.6.7520.04

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Memory Size	Gap	SSH	device memory in MB
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Count	Implemented	SSH	device CPU
	Storage Pool	Storage Pool Id	Implemented	SSH	
		Name	Implemented	SSH	
		Type	Gap	SSH	
		Thin Provisioning Supported	Implemented	SSH	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Include In Dwh Capacity	Implemented	SSH	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Dedupe Enabled	Implemented	SSH	Is dedupe enabled on the storage pool
		Virtual	Implemented	SSH	Is this a storage virtualization device?
		Raid Group	Implemented	SSH	indicates whether this storagePool is a raid group
		Snapshot Used Capacity	Implemented	SSH	
		Data Used Capacity	Implemented	SSH	
		Total Used Capacity	Implemented	SSH	Total capacity in MB
		Total Allocated Capacity	Implemented	SSH	
		Raw to Usable Ratio	Implemented	SSH	ratio to convert from usable capacity to raw capacity

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
HDS HNAS CLI	SSH	SSH	22		true	true	true	true

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HPE Nimble / Alletra 6000 Storage

Models and versions supported by this data collector:

API versions	Models	Firmware versions
v1	6030 AF1000 AF20Q AF3000 AF40 AF5000 CS1000 CS300 CS3000 CS500 CS5000 HF20 HF20H HF40 HF60	5.0.10.0-742719-opt 5.0.7.0-604814-opt 5.0.8.0-677726-opt 5.2.1.1000-1017822-opt 5.2.1.400-796142-opt 5.2.1.600-841103-opt 5.2.1.700-882343-opt 5.2.1.800-930936-opt 5.2.1.900-1003439-opt 6.0.0.300-956221-opt 6.0.0.400-991061-opt 6.1.1.200-1020304-opt 6.1.1.300-1028597-opt

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	WWN Alias	Host Aliases	Implemented	HTTPS	
		Object Type	Implemented	HTTPS	
		Source	Implemented	HTTPS	
		WWN	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
276					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
	Volume	Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		Total Compression Savings	Implemented	HTTPS	
		Compression Savings Space	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Latency Read	Implemented	HTTPS	
		Latency Total	Implemented	HTTPS	
		Latency Write	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
HP Nimble REST API	HTTPS	HTTPS	5392		true	false	true	true

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Huawei OceanStor (REST/HTTPS)

Models and versions supported by this data collector:

Models	Firmware versions
5300 V5 5500 V3 5500 V5 5800 V3 Dorado 5000 V6 SAS Dorado 6000 V3 Dorado 6000 V6 NVMe	V300R001C01 V300R002C10 V300R006C20 V300R006C50 V500R007C10 V500R007C30 V600R003C00 V600R005C03

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Volume	Capacity	Implemented	HTTPS	Snapshot Used capacity in MB
		Junction Path	Implemented	HTTPS	
		Name	Implemented	HTTPS	
		Total Raw Capacity	Implemented	HTTPS	Total raw capacity (sum of all disks on the array)
		Redundancy	Implemented	HTTPS	Redundancy level
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioned	Implemented	HTTPS	
		UUID	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
	Volume Map	LUN	Implemented	HTTPS	Name of the backend lun
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		IOPs Total	Implemented	HTTPS	
		Latency Write	Implemented	HTTPS	
		Utilization Total	Implemented	HTTPS	
	Volume	Cache Hit Ratio Read	Implemented	HTTPS	
		Cache Hit Ratio Total	Implemented	HTTPS	
		Cache Hit Ratio Write	Implemented	HTTPS	
		Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Latency Read	Implemented	HTTPS	
		Latency Total	Implemented	HTTPS	
		Latency Write	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Huawei OceanStor REST API	HTTPS	HTTPS	8088		true	true	true	true

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Huawei OceanStor Performance REST API	HTTPS	HTTPS	8088		true	false	true	true

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IBM Cleversafe

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
290					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		UUID	Implemented	HTTPS	
		Version	Implemented	HTTPS	software version
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Management Addresses	Implemented	HTTPS	
	Storage Pool	Include In Dwh Capacity	Implemented	HTTPS	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	HTTPS	
		Physical Disk Capacity (MB)	Implemented	HTTPS	used as raw capacity for storage pool
		Raid Group	Implemented	HTTPS	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	HTTPS	ratio to convert from usable capacity to raw capacity
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Total Used Capacity	Implemented	HTTPS	Total capacity in MB
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
IBM Cleversafe REST API	HTTPS	HTTPS	443		true	true	true	true

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IBM DS 8K (DSCLI)

Models and versions supported by this data collector:

Models	Firmware versions
2107-951 2107-961 2107-985 2107-996	7.6.31.4250 7.7.51.1400 7.8.57.18 7.9.21.91 7.9.32.126

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Type	Gap	DSNI	
		Virtual	Implemented	DSNI	Is this a storage virtualization device?
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Volume	Capacity	Implemented	DSNI	Snapshot Used capacity in MB
		Disk Type	Not Available	DSNI	
		Junction Path	Implemented	DSNI	
		Name	Implemented	Manual Entry	
		Total Raw Capacity	Implemented	DSNI	Total raw capacity (sum of all disks on the array)
		Storage Pool Id	Implemented	DSNI	
		Thin Provisioned	Implemented	DSNI	
		Type	Gap	DSNI	
		Used Capacity	Implemented	DSNI	
	Volume Map	LUN	Implemented	DSNI	Name of the backend lun
		Protocol Controller	Implemented	DSNI	
		Storage Port	Implemented	DSNI	
	Volume Mask	Initiator	Implemented	DSNI	
		Protocol Controller	Implemented	DSNI	
		Storage Port	Implemented	DSNI	
	WWN Alias	Host Aliases	Implemented	DSNI	
		Host OS	Implemented	DSNI	
		Object Type	Implemented	DSNI	
		Source	Implemented	DSNI	
		WWN	Implemented	DSNI	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Write	Implemented	DSNI	
		Read Utilization	Implemented	DSNI	
		Utilization Total	Implemented	DSNI	
		Utilization Write	Implemented	DSNI	
	Volume	Cache Hit Ratio Read	Implemented	DSNI	
		Cache Hit Ratio Total	Implemented	DSNI	
		Cache Hit Ratio Write	Implemented	DSNI	
		Raw Capacity	Implemented	DSNI	
		Total Capacity	Implemented	DSNI	
		IOps Read	Implemented	DSNI	Number of read IOps on the disk
		IOPs Total	Implemented	DSNI	
		IOPs Write	Implemented	DSNI	
		Latency Read	Implemented	DSNI	
		Latency Total	Implemented	DSNI	
		Latency Write	Implemented	DSNI	
		Throughput Read	Implemented	DSNI	
		Throughput Total	Implemented	DSNI	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	DSNI	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Data source wizard configuration	Manual Entry				true	true	true	true
IBM DS CLI	DSNI	DSNI			true	true	true	true

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IBM PowerVM (SSH)

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Name	Implemented	SSH	
		OID	Implemented	SSH	
		Type	Gap	SSH	
	VirtualMachine	Dns Name	Implemented	SSH	
		Guest State	Implemented	SSH	
		Host OID	Implemented	SSH	
		IPs	Implemented	SSH	
		MOID	Implemented	SSH	
		Memory	Implemented	SSH	
		Name	Implemented	SSH	
		OID	Implemented	SSH	
		OS	Implemented	SSH	
		Power State	Implemented	SSH	
		State Change Time	Implemented	SSH	
		Processors	Implemented	SSH	
	VirtualMachine Disk	OID	Implemented	SSH	
		VirtualDisk OID	Implemented	SSH	
		VirtualMachine OID	Implemented	SSH	
	Host	Host Cpu Count	Implemented	SSH	
		Host Installed Memory	Implemented	SSH	
		Host Model	Implemented	SSH	
		NIC count	Implemented	SSH	
		IPs	Implemented	SSH	
		Manufacturer	Implemented	SSH	
		Name	Implemented	SSH	
		OID	Implemented	SSH	
		Platform Type	Implemented	SSH	
	Info	DataSource Name	Implemented	SSH	Info
		Date	Implemented	SSH	
		Originator ID	Implemented	SSH	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
IBM Hardware Management Console SSH access	SSH	SSH	22		true	false	true	true

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IBM SVC (CLI)

Models and versions supported by this data collector:

Models	Firmware versions
2072-12F 2072-12G 2072-2N4 2072-324 2072-3H4 2072-3N4 2076-124 2076-12F 2076-224 2076-24F 2076-24G 2076-624 2076-724 2076-824 2076-AF6 2076-AFF 2077-24F 2077-424 2078-12F 2078-224 2078-24C 2078-24F 2078-324 2078-424 2078-4H4 2078-92G 2078-AF3 4657-924 4662-12G 4662-6H2 4666-AH8 9843-AE2 9843-AE3 9846-AG8 9848-AE2 9848-AF7 9848-AF8 9848-AG8 SVC	1.5.2.7 1.6.1.2 1.6.1.4 1.6.1.5 7.5.0.11 7.5.0.12 7.7.1.8 7.8.1.14 7.8.1.6 7.8.1.8 8.2.1.10 8.2.1.11 8.2.1.14 8.2.1.9 8.3.1.1 8.3.1.2 8.3.1.5 8.3.1.6 8.3.1.7 8.3.1.9 8.4.0.10 8.4.0.11 8.4.0.6 8.4.0.7 8.4.0.8 8.4.0.9 8.5.0.5 8.5.0.6 8.5.0.7 8.5.0.8 8.5.0.9 8.5.2.2 8.5.3.1 8.5.4.0

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					Host in MB
		Compression Enabled	Implemented	SSH	
		Encrypted	Implemented	SSH	
	Volume Map	LUN	Implemented	SSH	Name of the backend lun
		Protocol Controller	Implemented	SSH	
		Storage Port	Implemented	SSH	
	Volume Mask	Initiator	Implemented	SSH	
		Protocol Controller	Implemented	SSH	
		Storage Port	Implemented	SSH	
		Type	Gap	SSH	
	WWN Alias	Host Aliases	Implemented	SSH	
		Object Type	Implemented	SSH	
		Source	Implemented	SSH	
		WWN	Implemented	SSH	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Throughput Write	Implemented	SSH	
		Read Utilization	Implemented	SSH	
		Utilization Total	Implemented	SSH	
		Utilization Write	Implemented	SSH	
	Volume	Cache Hit Ratio Read	Implemented	SSH	
		Cache Hit Ratio Total	Implemented	SSH	
		Cache Hit Ratio Write	Implemented	SSH	
		Raw Capacity	Implemented	SSH	
		Total Capacity	Implemented	SSH	
		Used Capacity	Implemented	SSH	
		Written Capacity	Implemented	SSH	
		Capacity Used Ratio	Implemented	SSH	
		CapacityRatio Written	Implemented	SSH	
		IOps Read	Implemented	SSH	Number of read IOps on the disk
		IOPs Total	Implemented	SSH	
		IOPs Write	Implemented	SSH	
		Latency Read	Implemented	SSH	
		Latency Total	Implemented	SSH	
		Latency Write	Implemented	SSH	
		Throughput Read	Implemented	SSH	
		Throughput Total	Implemented	SSH	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	SSH	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
IBM SVC CLI	SSH	SSH	22		true	false	true	true

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IBM XIV & A9000 (XIVCLI)

Models and versions supported by this data collector:

Models	Firmware versions
415 A14	10.2.4.e 12.3.2.c

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Type	Gap	XIV CLI	
		Virtual	Implemented	XIV CLI	Is this a storage virtualization device?
	Volume	Capacity	Implemented	XIV CLI	Snapshot Used capacity in MB
		DiskGroup	Implemented	XIV CLI	Disk Group Type
		Disk Type	Not Available	XIV CLI	
		Name	Implemented	XIV CLI	
		Qtree Id	Implemented	XIV CLI	unique id of the qtree
		Total Raw Capacity	Implemented	XIV CLI	Total raw capacity (sum of all disks on the array)
		Redundancy	Implemented	XIV CLI	Redundancy level
		Storage Pool Id	Implemented	XIV CLI	
		Thin Provisioned	Implemented	XIV CLI	
		Type	Gap	XIV CLI	
		Used Capacity	Implemented	XIV CLI	
		Compression Enabled	Implemented	XIV CLI	
	Volume Map	LUN	Implemented	XIV CLI	Name of the backend lun
		Protocol Controller	Implemented	XIV CLI	
	Volume Mask	Initiator	Implemented	XIV CLI	
		Protocol Controller	Implemented	XIV CLI	
	WWN Alias	Host Aliases	Implemented	XIV CLI	
		Host OS	Implemented	XIV CLI	
		Object Type	Implemented	XIV CLI	
		Source	Implemented	XIV CLI	
		WWN	Implemented	XIV CLI	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

		Used Capacity	Implemented	DSNI	
		Raw Capacity	Implemented	DSNI	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional information**
		Capacity Ratio	Implemented	DSNI	Reported in a time series
	Volume	Latency Total	Implemented	DSNI	
		Latency Read	Implemented	DSNI	
		IOPs Write	Implemented	DSNI	
		Compression Savings Space	Implemented	DSNI	
		Throughput Read	Implemented	DSNI	
		IOPs Total	Implemented	DSNI	
		Latency Write	Implemented	DSNI	
		IOps Read	Implemented	DSNI	Number of read IOps on the disk
		Cache Hit Ratio Read	Implemented	DSNI	
		Total Compression Savings	Implemented	DSNI	
		Cache Hit Ratio Total	Implemented	DSNI	
		Cache Hit Ratio Write	Implemented	DSNI	
		Throughput Write	Implemented	DSNI	
		Throughput Total	Implemented	DSNI	Average disk total rate (read and write across all disks) in MB/s

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
IBM DS CLI	DSNI	DSNI			true	true	true	true
IBM XIV CLI	XIV CLI	TCP	7778		true	false	true	false

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Infinidat Infinibox (HTTP)

Models and versions supported by this data collector:

Models	Firmware versions
F6230 F6240 F6303 F6304	6.0.31.0 7.0.14.20

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Protocol Controller	Implemented	HTTPS	
		Type	Gap	HTTPS	
		Storage Port	Implemented	HTTPS	
	WWN Alias	Source	Implemented	HTTPS	
		Host Aliases	Implemented	HTTPS	
		WWN	Implemented	HTTPS	
		Object Type	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Infinidat REST API	HTTPS	HTTPS	443		true	true	true	true

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Microsoft Azure Compute

Models and versions supported by this data collector:

API versions
2018-06-01

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Manufacturer	Implemented	HTTPS	
		Name	Implemented	HTTPS	
		OID	Implemented	HTTPS	
	Info	Api Description	Implemented	HTTPS	
		Api Name	Implemented	HTTPS	
		Api Version	Implemented	HTTPS	
		DataSource Name	Implemented	HTTPS	Info
		Date	Implemented	HTTPS	
		Originator ID	Implemented	HTTPS	
		Originator Key	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance	Data Store	Capacity Provisioned	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Over Commit Capacity Ratio	Implemented	HTTPS	Reported as a time series
	Virtual Disk	Total Capacity	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
	vm	Total CPU Utilization	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		diskIops.total	Implemented	HTTPS	
		Disk IOPs write	Implemented	HTTPS	
		Disk Throughput Read	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	total disk throughput read
		Disk Throughput Write	Implemented	HTTPS	
		IP Throughput Read	Implemented	HTTPS	
		Throughput total	Implemented	HTTPS	IP throughput total
		ipThroughput.write	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Microsoft Azure Compute REST API	HTTPS	HTTPS	443		true	true	true	true

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Microsoft Hyper-V

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
340					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Manufacturer	Implemented	WMI	
		Name	Implemented	WMI	
		OID	Implemented	WMI	
		PlatformType	Implemented	WMI	
	ISCSI Node	Host Aliases	Implemented	WMI	
		Node Name	Implemented	WMI	
		OID	Implemented	WMI	
		Type	Gap	WMI	
	Info	DataSource Name	Implemented	WMI	Info
		Date	Implemented	WMI	
		Originator ID	Implemented	WMI	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

				Management	total rate (read and write across all disks) in MB/s
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Total Capacity	Implemented	WS-Management	
	vm	Total Capacity	Implemented	WS-Management	
		Used Capacity	Implemented	WS-Management	
		Capacity Used Ratio	Implemented	WS-Management	
		Total CPU Utilization	Implemented	WS-Management	
		IOps Read	Implemented	WS-Management	Number of read IOps on the disk
		diskIops.total	Implemented	WS-Management	
		Disk IOPs write	Implemented	WS-Management	
		Latency Total	Implemented	WS-Management	
		Disk Throughput Read	Implemented	WS-Management	
		Throughput Read	Implemented	WS-Management	total disk throughput read
		Disk Throughput Write	Implemented	WS-Management	
		IP Throughput Read	Implemented	WS-Management	
		Throughput total	Implemented	WS-Management	IP throughput total
		ipThroughput.write	Implemented	WS-Management	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
PowerShell	WS-Management	HTTP	5985		true	false	false	true
WMI	WMI	WMI	135		true	false	true	true

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NetApp 7 Mode

Models and versions supported by this data collector:

API versions	Models	Firmware versions
1.12 1.14 1.17 1.19 1.20 1.21	FAS2040 FAS2050 FAS2220 FAS2240-2 FAS2240-4 FAS2520 FAS2554 FAS3140 FAS3160 FAS3210 FAS3220 FAS3240 FAS3250 FAS3270 FAS6240 FAS6290 FAS8020 FAS8040 FAS8060 FAS8080 N6070 N6240 V3240	7.3.6 8.1.1 7-Mode 8.1.3P2 7-Mode 8.1.4P1 7-Mode 8.1.4P10 7-Mode 8.1.4P9D18 7-Mode 8.2.1 7-Mode 8.2.2 7-Mode 8.2.3 7-Mode 8.2.3P2 7-Mode 8.2.3P3 7-Mode 8.2.4 7-Mode 8.2.4P2 7-Mode 8.2.4P4 7-Mode 8.2.4P5 7-Mode 8.2.4P6 7-Mode 8.2.5 7-Mode 8.2.5P1 7-Mode 8.2.5P2 7-Mode 8.2.5P4 7-Mode 8.2.5P5 7-Mode 8.2P3 7-Mode 8.2P4 7-Mode Data ONTAP Release 7.3.3 Data ONTAP Release 7.3.4 Data ONTAP Release 8.2.5 7-Mode

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
348					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

					backend lun
		Protocol Controller	Implemented		
		Type	Gap		
	Volume Mask	Initiator	Implemented		
		Protocol Controller	Implemented		
		Storage Port	Implemented		
		Type	Gap		

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					
360					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Write	Implemented		
		Read Utilization	Implemented		
		Utilization Total	Implemented		
		Utilization Write	Implemented		
	Volume	Raw Capacity	Implemented		
		Total Capacity	Implemented		
		Used Capacity	Implemented		
		Capacity Used Ratio	Implemented		
		IO Density Read	Implemented		
		IO Density Total	Implemented		
		Write IO Density	Implemented		
		IOps Read	Implemented		Number of read IOps on the disk
		IOPs Total	Implemented		
		IOPs Write	Implemented		
		Latency Read	Implemented		
		Latency Total	Implemented		
		Latency Write	Implemented		
		Partial Blocked Ratio	Implemented		
		Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
		Write Pending	Implemented		total write pending

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
NetApp 7 Mode ZAPI	ZAPI	ZAPI			true	true	true	true

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NetApp Cloud Volumes Service

Models and versions supported by this data collector:

Models	Firmware versions
AWS Cloud Volumes	v1

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
366					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
					are spare)
		Failed Raw Capacity	Implemented		Raw capapcity of failed disks (sum of all disks that are failed)
	Storage Pool	Storage Pool Id	Implemented		
		Name	Implemented		
		Type	Gap		
		Thin Provisioning Supported	Implemented		Whether this internal volume supports thin provisioning for the volume layer on top of it
		Include In Dwh Capacity	Implemented		A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Virtual	Implemented		Is this a storage virtualization device?
		Raid Group	Implemented		indicates whether this storagePool is a raid group
		Snapshot Used Capacity	Implemented		
		Data Used Capacity	Implemented		
		Data Allocated Capacity	Gap		capacity allocated for data
		Total Used Capacity	Implemented		Total capacity in MB
		Total Allocated Capacity	Implemented		
		Physical Disk Capacity (MB)	Implemented		used as raw capacity for storage pool
		Raw to Usable Ratio	Implemented		ratio to convert from usable capacity to raw capacity

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Cloud Volumes Service REST API	HTTPS	HTTPS	443		true	true	true	true

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Amazon FSx for NetApp ONTAP

Models and versions supported by this data collector:

Models	Firmware versions
FSx for ONTAP	Data ONTAP

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

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Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

	VirtualMachine StoragePool IRef	Storage Pool Id	Implemented	HTTPS	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	VirtualMachine Volume Ref	Volume	Implemented	HTTPS	
	Volume	Capacity	Implemented	HTTPS	Snapshot Used capacity in MB
		DiskGroup	Implemented	HTTPS	Disk Group Type
		Junction Path	Implemented	HTTPS	
		Last Known Access Time	Implemented	HTTPS	Last know access to the volume
		Name	Implemented	HTTPS	
		Qtree Id	Implemented	HTTPS	unique id of the qtree
		Total Raw Capacity	Implemented	HTTPS	Total raw capacity (sum of all disks on the array)
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioned	Implemented	HTTPS	
		Type	Gap	HTTPS	
		UUID	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Encrypted	Implemented	HTTPS	
	Volume Map	LUN	Implemented	HTTPS	Name of the backend lun
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
performance	Qtree.1+	Storage	Failed Disks	Implemented	HTTPS
		Storage Node	Cache Hit Ratio Total	Implemented	HTTPS
			Total Disk Read Replaced	Implemented	HTTPS
			Utilization Total	Implemented	HTTPS

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
NetApp ONTAP API	HTTP/HTTPS	HTTP/HTTPS	80/443		true	true	true	true

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NetApp Clustered Data ONTAP 8.1.1+

Models and versions supported by this data collector:

Models	Firmware versions
AFF-A150 AFF-A200 AFF-A220 AFF-A250 AFF-A300 AFF-A320 AFF-A400 AFF-A700 AFF-A700s AFF-A800 AFF-A900 AFF-C190 AFF-C250 AFF-C400 AFF-C800 AFF8020 AFF8040 AFF8060 AFF8080 CDvM100 CDvM200 DM5000H FAS2240-2 FAS2240-4 FAS2520 FAS2552 FAS2554 FAS2620 FAS2650 FAS2720 FAS2750 FAS3220 FAS3250 FAS3270 FAS500f FAS6210 FAS6220 FAS8020 FAS8040 FAS8060 FAS8080 FAS8200 FAS8300 FAS8700 FAS9000 FAS9500 FASDvM300 SIMBOX V6240	8.2.3P5 8.3.0 8.3.1 8.3.1P2 8.3.2 8.3.2P12 8.3.2P2 8.3.2P5 9.0.1 9.1.0 9.1.0P1 9.1.0P10 9.1.0P11 9.1.0P12 9.1.0P14 9.1.0P15 9.1.0P17 9.1.0P19 9.1.0P20 9.1.0P5 9.1.0P7 9.1.0P8 9.10.0 9.10.1 9.10.1P1 9.10.1P10 9.10.1P11 9.10.1P12 9.10.1P13 9.10.1P2 9.10.1P3 9.10.1P4 9.10.1P5 9.10.1P6 9.10.1P7 9.10.1P8 9.10.1P9 9.11.0P1 9.11.1 9.11.1P1 9.11.1P10 9.11.1P2 9.11.1P3 9.11.1P4 9.11.1P5 9.11.1P6 9.11.1P7 9.11.1P8 9.11.1P9 9.11.1X12 9.11.1X26 9.12.1 9.12.1P1 9.12.1P2

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

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Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Qos Limit IOPS	Implemented	HTTPS	
		Qos Limit MBPS	Implemented	HTTPS	
		Qos Limit Raw	Implemented	HTTPS	
		QoS - Policy	Implemented	HTTPS	
	Volume Map	LUN	Implemented	HTTPS	Name of the backend lun
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Throughput Write	Implemented	HTTPS	
	Volume	Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		IO Density Read	Implemented	HTTPS	
		IO Density Total	Implemented	HTTPS	
		Write IO Density	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Latency Read	Implemented	HTTPS	
		Latency Total	Implemented	HTTPS	
		Latency Write	Implemented	HTTPS	
		Partial Blocked Ratio	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
NetApp ONTAP API	HTTP/HTTPS	HTTP/HTTPS	80/443		true	true	true	true

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NetApp SolidFire 8.1+

Models and versions supported by this data collector:

Models	Firmware versions
H410S-2 H610S-2 H610S-4 SF19210 SF2405 SF38410 SF4805 SF9605 SF9608 FCN001 H300S H410S-0 H410S-1 H410S-2 H500S H610S-1 H610S-2 H610S-4 H610S2 SF19210 SF38410 SF4805 SF9605	11.1.0.72 11.5.0.63 11.7.0.76 11.8.0.23 12.0.0.333 12.2.0.777 12.3.0.958 12.3.1.103 12.3.1.165 12.3.2.3 12.5.0.897 12.7.0.380

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
418					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Qos Burst IOPS	Implemented	HTTPS	
		Qos Limit IOPS	Implemented	HTTPS	
		qos Min IOPS	Implemented	HTTPS	
		QoS - Policy	Implemented	HTTPS	
	Volume Map	LUN	Implemented	HTTPS	Name of the backend lun
		Masking Required	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Other Total Capacity	Implemented	HTTPS	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Capacity	Implemented	HTTPS	
	Volume	Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		Total Compression Savings	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Latency Read	Implemented	HTTPS	
		Latency Total	Implemented	HTTPS	
		Latency Write	Implemented	HTTPS	
		Partial Blocked Ratio	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
		Utilization Total	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
SolidFire REST API	HTTPS	HTTPS	443		true	true	true	true

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NetApp StorageGRID (HTTPS)

Models and versions supported by this data collector:

API versions	Models	Firmware versions
3.0 3.2 3.3 3.4 3.5	Webscale	11.2.0 11.4.0 11.4.0.3 11.4.0.4 11.5.0.1 11.5.0.11 11.5.0.2 11.5.0.3 11.5.0.6 11.5.0.7 11.5.0.8 11.5.0.9 11.6.0 11.6.0.1 11.6.0.10 11.6.0.2 11.6.0.4 11.6.0.5 11.6.0.7 11.6.0.8 11.6.0.9 11.7.0

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Node Capacity Utilization Used Meta Data in MB	Implemented	HTTPS	
		Site Name	Implemented	HTTPS	
	Storage Pool	Include In Dwh Capacity	Implemented	HTTPS	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	HTTPS	
		Physical Disk Capacity (MB)	Implemented	HTTPS	used as raw capacity for storage pool
		Raid Group	Implemented	HTTPS	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	HTTPS	ratio to convert from usable capacity to raw capacity
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Total Used Capacity	Implemented	HTTPS	Total capacity in MB
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

	Implemented				Throughput Total
	Implemented		Average disk total rate (read and write across all disks) in MB/s		Throughput Write
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Implemented			StoragePool Disk	Capacity Provisioned
	Implemented				Raw Capacity
	Implemented				Total Capacity
	Implemented				Used Capacity
	Implemented				Over Commit Capacity Ratio
	Implemented		Reported as a time series		Capacity Used Ratio

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
StorageGrid REST API	HTTPS	HTTPS	443		true	false	true	true

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Nutanix Storage (REST)

Models and versions supported by this data collector:

Models	Firmware versions
HPE DL360-8 G10 NX-3060-G6 NX-3170-G6 NX-8035-G6 NX-8150-G7 HPE DL360-8 G10 HPE DL380-12 G10 NX-3060-G5 NX-3170-G7 NX-5155-G6 NX-8035-G6 NX-8035-G7 NX-8150-G7 NX-8150-G8	6.5.1.6 6.5.2 6.5.2.5 6.5.2.6 6.5.2.7 6.5.3 6.5.3.1

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
440					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Volume	Capacity	Implemented	HTTPS	Snapshot Used capacity in MB
		Junction Path	Implemented	HTTPS	
		Name	Implemented	HTTPS	
		Qtree Id	Implemented	HTTPS	unique id of the qtree
		Total Raw Capacity	Implemented	HTTPS	Total raw capacity (sum of all disks on the array)
		Redundancy	Implemented	HTTPS	Redundancy level
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioned	Implemented	HTTPS	
		UUID	Implemented	HTTPS	
	Volume Map	LUN	Implemented	HTTPS	Name of the backend lun
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Volume Mask	Initiator	Implemented	HTTPS	
		Protocol Controller	Implemented	HTTPS	
		Storage Port	Implemented	HTTPS	
		Type	Gap	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

	Implemented	HTTPS			Throughput Total
	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s		Throughput Write
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
	Implemented	HTTPS		Volume	IOps Read
	Implemented	HTTPS	Number of read IOps on the disk		IOPs Total
	Implemented	HTTPS			IOPs Write
	Implemented	HTTPS			Latency Read
	Implemented	HTTPS			Latency Total
	Implemented	HTTPS			Latency Write
	Implemented	HTTPS			Throughput Read
	Implemented	HTTPS			Throughput Total
	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s		Throughput Write

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Nutanix REST API	HTTPS	HTTPS	443		true	true	true	true

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OpenStack (REST API / SSH)

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Node Name	Implemented	HTTPS	
		OID	Implemented	HTTPS	
		Type	Gap	HTTPS	
	Info	DataSource Name	Implemented	HTTPS	Info
		Date	Implemented	HTTPS	
		Originator ID	Implemented	HTTPS	
		Originator Key	Implemented	HTTPS	
performance	Data Store	Total Capacity	Implemented		
		Capacity Used Ratio	Implemented		
		Capacity Provisioned	Implemented		
		Used Capacity	Implemented		
		Over Commit Capacity Ratio	Implemented		Reported as a time series
	Host	Total CPU Utilization	Implemented		
		Total Memory Utilization	Implemented		
	Virtual Disk	Latency Read	Implemented		
		Latency Total	Implemented		
		Latency Write	Implemented		

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
OpenStack REST API	HTTPS	HTTPS	443		true	false	true	true
OpenStack SSH	SSH	SSH	22		true	false	true	true

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Oracle ZFS (HTTPS)

Models and versions supported by this data collector:

Models	Firmware versions
Sun ZFS Storage 7330 Sun ZFS Storage 7335 Sun ZFS Storage 7350 Sun ZFS Storage 7370 Sun ZFS Storage 7420 Sun ZFS Storage 7430 Sun ZFS Storage 7450	1-1.1 1-1.2 1-1.3 1-1.34 1-1.4 2013.06.05.6.12 2013.06.05.6.15 2013.06.05.7.21 2013.06.05.7.24 2013.06.05.7.25 2013.06.05.7.26 2013.06.05.8.0 2013.06.05.8.26 2013.06.05.8.29 2013.06.05.8.35 2013.06.05.8.37 2013.06.05.8.47 2013.06.05.8.50 2013.06.05.8.53 2013.06.05.8.6 2013.06.05.8.7

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Required			
		Protocol Controller	Implemented	HTTP/S	
		Type	Gap	HTTP/S	
	Volume Mask	Storage Port	Implemented	HTTP/S	
		Initiator	Implemented	HTTP/S	
		Protocol Controller	Implemented	HTTP/S	
		Type	Gap	HTTP/S	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

					and write across all disks) in MB/s
		Throughput Write	Implemented		
		Utilization Total	Implemented		

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
	Storage Node Data	key	Implemented		
		Server ID	Implemented		
		Throughput Read	Implemented		
		Throughput Write	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		IOps Read	Implemented		Number of read IOps on the disk
		IOPs Write	Implemented		
		IOPs Total	Implemented		
		Cache Hit Ratio Total	Implemented		
		Utilization Total	Implemented		
	StoragePool Disk	IOPs Total	Implemented		
		Total Capacity	Implemented		
		Capacity Used Ratio	Implemented		
		Total Data Capacity	Implemented		
		Capacity Provisioned	Implemented		
		Data Used Capacity	Implemented		
		Used Capacity	Implemented		
		Other Used Capacity	Implemented		
		Raw Capacity	Implemented		
		Over Commit Capacity Ratio	Implemented		Reported as a time series
		Snapshot Used Capacity	Implemented		
		Snapshot Used Capacity Ratio	Implemented		Reported as a time series

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Oracle ZFS REST API	HTTP/HTTPS	HTTP/HTTPS	215		true	true	true	true

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Pure Storage FlashArray (HTTP)

Models and versions supported by this data collector:

Models	Firmware versions
DFSC1 FA-420 FA-450 FA-C40R3 FA-C60 FA-C60R3 FA-X10R2 FA-X10R3 FA-X20R2 FA-X20R3 FA-X50R2 FA-X50R3 FA-X70R2 FA-X70R3 FA-X90R2 FA-X90R3 FA-XL130 FA-XL170 FA-m10r2 FA-m20 FA-m20r2 FA-m50 FA-m50r2 FA-m70 FA-m70r2 FA-x70	4.8.8 5.3.14 5.3.15 5.3.17 5.3.18 5.3.20 5.3.21 5.3.6 5.3.8 6.1.10 6.1.11 6.1.13 6.1.14 6.1.15 6.1.17 6.1.18 6.1.19 6.1.21 6.1.22 6.1.23 6.1.5 6.2.13 6.2.7 6.2.9 6.3.10 6.3.11 6.3.12 6.3.2 6.3.5 6.3.6 6.3.7 6.3.9 6.4.3 6.4.4 6.4.5

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Protocol Controller	Implemented	HTTP/S	
		Storage Port	Implemented	HTTP/S	
		Type	Gap	HTTP/S	
	Volume Mask	Initiator	Implemented	HTTP/S	
		Protocol Controller	Implemented	HTTP/S	
		Storage Port	Implemented	HTTP/S	
		Type	Gap	HTTP/S	
	WWN Alias	Host Aliases	Implemented	HTTP/S	
		Object Type	Implemented	HTTP/S	
		Source	Implemented	HTTP/S	
		WWN	Implemented	HTTP/S	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		
	StoragePool Disk	Capacity Provisioned	Implemented		
		Raw Capacity	Implemented		
		Total Capacity	Implemented		
		Used Capacity	Implemented		
		Over Commit Capacity Ratio	Implemented		Reported as a time series
		Capacity Used Ratio	Implemented		
		Total Data Capacity	Implemented		
		Data Used Capacity	Implemented		
		Other Used Capacity	Implemented		
		Snapshot Used Capacity	Implemented		
		Snapshot Used Capacity Ratio	Implemented		Reported as a time series
	Volume	Raw Capacity	Implemented		
		Total Capacity	Implemented		
		Used Capacity	Implemented		
		Capacity Used Ratio	Implemented		
		IOps Read	Implemented		Number of read IOps on the disk
		IOPs Total	Implemented		
		IOPs Write	Implemented		
		Latency Read	Implemented		
		Latency Total	Implemented		
		Latency Write	Implemented		
		Throughput Read	Implemented		
		Throughput Total	Implemented		Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented		

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Pure Storage REST API	HTTP/HTTPS	HTTP/HTTPS	80/443		true	true	true	true

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Red Hat RHV (REST)

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

	VirtualMachine Disk	OID	Implemented	HTTP/S	
		VirtualMachine OID	Implemented	HTTP/S	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		VirtualDisk OID	Implemented	HTTP/S	
	Host	OID	Implemented	HTTP/S	
		Name	Implemented	HTTP/S	
		IPs	Implemented	HTTP/S	
		Platform Type	Implemented	HTTP/S	
		Host Installed Memory	Implemented	HTTP/S	
		Manufacturer	Implemented	HTTP/S	
		Host Model	Implemented	HTTP/S	
		Host Cpu Count	Implemented	HTTP/S	
		Host Cpu Speed	Implemented	HTTP/S	
		NIC count	Implemented	HTTP/S	
		NIC speed	Implemented	HTTP/S	
	ISCSI Node	OID	Implemented	HTTP/S	
		Node Name	Implemented	HTTP/S	
		Type	Gap	HTTP/S	
	Info	DataSource Name	Implemented	HTTP/S	Info
		Originator ID	Implemented	HTTP/S	
		Date	Implemented	HTTP/S	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Red Hat RHEV REST API	HTTPS	HTTPS	443		true	true	true	true

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Rubrik Storage

Models and versions supported by this data collector:

API versions	Firmware versions
v5.3	5.3.3-p1-19391 6.0.3-p3-13584 7.0.2-p4-15876 7.0.3-p1-15949 8.0.3-p2-22743

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		Node Capacity Utilization Used in MB	Implemented	HTTPS	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		Site Name	Implemented	HTTPS	
	Storage Pool	Data Used Capacity	Implemented	HTTPS	
		Include In Dwh Capacity	Implemented	HTTPS	A way from ACQ to cotnrol which stroage pools are interesting in DWH Capacity
		Name	Implemented	HTTPS	
		Other UsedCapacity (MB)	Implemented	HTTPS	Any capacity other than data and snapshot
		Physical Disk Capacity (MB)	Implemented	HTTPS	used as raw capacity for storage pool
		Raid Group	Implemented	HTTPS	indicates whether this storagePool is a raid group
		Raw to Usable Ratio	Implemented	HTTPS	ratio to convert from usable capacity to raw capacity
		Snapshot Used Capacity	Implemented	HTTPS	
		Storage Pool Id	Implemented	HTTPS	
		Thin Provisioning Supported	Implemented	HTTPS	Whether this internal volume supports thin provisioning for the volume layer on top of it
		Total Allocated Capacity	Implemented	HTTPS	
		Total Used Capacity	Implemented	HTTPS	Total capacity in MB
		Type	Gap	HTTPS	
		Virtual	Implemented	HTTPS	Is this a storage virtualization device?
		Effective Used Capacity Percent	Implemented	HTTPS	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

					and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	
		Utilization Total	Implemented	HTTPS	
	StoragePool Disk	Raw Capacity	Implemented	HTTPS	
		Total Capacity	Implemented	HTTPS	
		Used Capacity	Implemented	HTTPS	
		Capacity Used Ratio	Implemented	HTTPS	
		Data Used Capacity	Implemented	HTTPS	
		IOps Read	Implemented	HTTPS	Number of read IOps on the disk
		IOPs Total	Implemented	HTTPS	
		IOPs Write	Implemented	HTTPS	
		Other Used Capacity	Implemented	HTTPS	
		Snapshot Used Capacity	Implemented	HTTPS	
		Throughput Read	Implemented	HTTPS	
		Throughput Total	Implemented	HTTPS	Average disk total rate (read and write across all disks) in MB/s
		Throughput Write	Implemented	HTTPS	

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
Rubrik Storage REST API	HTTPS	HTTPS	443		true	true	true	true

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NetApp HCI Virtual Center

Models and versions supported by this data collector:

API versions
VMware vCenter Server 6.7.0 build-10244857 VMware vCenter Server 6.7.0 build-14368073 VMware vCenter Server 7.0.3 build-19234570 VMware vCenter Server 7.0.3 build-20150588 VMware vCenter Server 7.0.3 build-20395099 VMware vCenter Server 7.0.3 build-20990077 VMware vCenter Server 7.0.3 build-21477706 VMware vCenter Server 7.0.3 build-21784236 VMware vCenter Server 8.0.1 build-21815093

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
490					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Manufacturer	Implemented	Web Services	
		Name	Implemented	Web Services	
		OID	Implemented	Web Services	
		PlatformeType	Implemented	Web Services	
	ISCSI Node	Host Aliases	Implemented	Web Services	
		Node Name	Implemented	Web Services	
		OID	Implemented	Web Services	
		Type	Gap	Web Services	
	Info	Api Description	Implemented	Web Services	
		Api Name	Implemented	Web Services	
		Api Version	Implemented	Web Services	
		Client Api Name	Implemented	Web Services	
		Client Api Version	Implemented	Web Services	
		DataSource Name	Implemented	Web Services	Info
		Date	Implemented	Web Services	
		Originator ID	Implemented	Web Services	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
		swapRate.inRate	Implemented	Web Services	
		Swap Rate	Implemented	Web Services	
		Total Swap Rate	Implemented	Web Services	
	vm	cpuCoSchedulingDelayTimePercent.total	Implemented	Web Services	
		cpuDemandToEntitlementPercent.total	Implemented	Web Services	
		Idle CPU Time	Implemented	Web Services	idle time in percent
		CPU Wait Time	Implemented	Web Services	total cpu wait time in percent
		Total CPU Utilization	Implemented	Web Services	
		IOps Read	Implemented	Web Services	Number of read IOps on the disk
		diskIops.total	Implemented	Web Services	
		Disk IOPs write	Implemented	Web Services	
		Latency Read	Implemented	Web Services	
		Latency Total	Implemented	Web Services	
		Latency Write	Implemented	Web Services	
		Disk Throughput Read	Implemented	Web Services	
		Throughput Read	Implemented	Web Services	total disk throughput read
		Disk Throughput Write	Implemented	Web Services	
		IP Throughput Read	Implemented	Web Services	
		Throughput total	Implemented	Web Services	IP throughput total
		ipThroughput.write	Implemented	Web Services	
		Total Memory Utilization	Implemented	Web Services	
		swapRate.inRate	Implemented	Web Services	
		Swap Rate	Implemented	Web Services	
		Total Swap Rate	Implemented	Web Services	
		Schedule wait time	Implemented	Web Services	Waiting to be scheduled time in percent

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
VMware REST API	Web Services	HTTP/HTTPS	80/443		true	true	true	true

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VMware Cloud on AWS

Models and versions supported by this data collector:

API versions
VMware vCenter Server 7.0.3 build-20532039 VMware vCenter Server 7.0.3 build-20870699 VMware vCenter Server 8.0.0 build-21709157

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					
498					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Host Cpu Count	Implemented	Web Services	
		Host Cpu Speed	Implemented	Web Services	
		NIC count	Implemented	Web Services	
		NIC speed	Implemented	Web Services	
	Info	DataSource Name	Implemented	Web Services	Info
		Originator ID	Implemented	Web Services	
		Date	Implemented	Web Services	
		Api Name	Implemented	Web Services	
		Api Version	Implemented	Web Services	
		Api Description	Implemented	Web Services	
		Client Api Name	Implemented	Web Services	
		Client Api Version	Implemented	Web Services	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
					and write across all disks) in MB/s
		IOPs Total	Implemented	Web Services	
		Latency Write	Implemented	Web Services	
	vm	Latency Write	Implemented	Web Services	
		Throughput total	Implemented	Web Services	IP throughput total
		Swap Rate	Implemented	Web Services	
		Disk IOPs write	Implemented	Web Services	
		cpuCoSchedulingDelayTimePercent.total	Implemented	Web Services	
		IOps Read	Implemented	Web Services	Number of read IOps on the disk
		Total Memory Utilization	Implemented	Web Services	
		Idle CPU Time	Implemented	Web Services	idle time in percent
		Latency Read	Implemented	Web Services	
		cpuDemandToEntitlementPercent.total	Implemented	Web Services	
		Latency Total	Implemented	Web Services	
		ipThroughput.write	Implemented	Web Services	
		IP Throughput Read	Implemented	Web Services	
		Disk Throughput Read	Implemented	Web Services	
		CPU Wait Time	Implemented	Web Services	total cpu wait time in percent
		swapRate.inRate	Implemented	Web Services	
		Disk Throughput Write	Implemented	Web Services	
		Total CPU Utilization	Implemented	Web Services	
		Schedule wait time	Implemented	Web Services	Waiting to be scheduled time in percent
		diskIops.total	Implemented	Web Services	
		Total Swap Rate	Implemented	Web Services	
		Throughput Read	Implemented	Web Services	total disk throughput read

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
VMware REST API	Web Services	HTTP/HTTPS	80/443		true	true	true	true

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VMware vSphere (Web Services)

Models and versions supported by this data collector:

API versions
VMware ESXi 6.0.0 build-10719132 VMware ESXi 6.0.0 build-2494585 VMware ESXi 6.0.0 build-5572656 VMware ESXi 6.0.0 build-9313334 VMware ESXi 6.5.0 build-14990892 VMware ESXi 6.5.0 build-5969303 VMware ESXi 7.0.0 build-15843807 VMware ESXi 7.0.3 build-20036589 VMware ESXi 7.0.3 build-20328353 VMware ESXi 7.0.3 build-20842708 VMware vCenter Server 5.0.0 build-3073236 VMware vCenter Server 5.0.0 build-455964 VMware vCenter Server 5.0.0 build-623373 VMware vCenter Server 5.1.0 build-3814779 VMware vCenter Server 5.5.0 build-1750787 VMware vCenter Server 5.5.0 build-2442329 VMware vCenter Server 5.5.0 build-3000241 VMware vCenter Server 5.5.0 build-3252642 VMware vCenter Server 5.5.0 build-3721164 VMware vCenter Server 5.5.0 build-4180647 VMware vCenter Server 5.5.0 build-6516310 VMware vCenter Server 5.5.0 build-9911218 VMware vCenter Server 6.0.0 build-13638472 VMware vCenter Server 6.0.0 build-14510545 VMware vCenter Server 6.0.0 build-2776511 VMware vCenter Server 6.0.0 build-3634793 VMware vCenter Server 6.0.0 build-3634794 VMware vCenter Server 6.0.0 build-5960847 VMware vCenter Server 6.0.0 build-7924803 VMware vCenter Server 6.0.0 build-8803875 VMware vCenter Server 6.0.0 build-9313458 VMware vCenter Server 6.5.0 build-10964411 VMware vCenter Server 6.5.0 build-15679215 VMware vCenter Server 6.5.0 build-17590285 VMware vCenter Server 6.5.0 build-17994927 VMware vCenter Server 6.5.0 build-18499837 VMware vCenter Server 6.5.0 build-18711281 VMware vCenter Server 6.5.0 build-19261680 VMware vCenter Server 6.5.0 build-20510539 VMware vCenter Server 6.5.0 build-7119157 VMware vCenter Server 6.7.0 build-10244857 VMware vCenter Server 6.7.0 build-11727113 VMware vCenter Server 6.7.0 build-13007421 VMware vCenter Server 6.7.0 build-13639324 VMware vCenter Server 6.7.0 build-14368073 VMware vCenter Server 6.7.0 build-15129973 VMware vCenter Server 6.7.0 build-15679289 VMware vCenter Server 6.7.0 build-17137327 VMware vCenter Server 6.7.0 build-18010599 VMware vCenter Server 6.7.0 build-18485185 VMware vCenter Server 6.7.0 build-18831049 VMware vCenter Server 6.7.0 build-19299595 VMware vCenter Server 6.7.0 build-19832247 VMware vCenter Server 6.7.0 build-19832280

Products supported by this data collector:

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
foundation					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribut	Status	Protocol Used	Additional Information
		Manufacturer	Implemented	Web Services	
		Name	Implemented	Web Services	
		OID	Implemented	Web Services	
		PlatformeType	Implemented	Web Services	
	ISCSI Node	Host Aliases	Implemented	Web Services	
		Node Name	Implemented	Web Services	
		OID	Implemented	Web Services	
		Type	Gap	Web Services	
	Info	Api Description	Implemented	Web Services	
		Api Name	Implemented	Web Services	
		Api Version	Implemented	Web Services	
		Client Api Name	Implemented	Web Services	
		Client Api Version	Implemented	Web Services	
		DataSource Name	Implemented	Web Services	Info
		Date	Implemented	Web Services	
		Originator ID	Implemented	Web Services	

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information
performance					

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

Product	Category	Feature/Attribute	Status	Protocol Used	Additional Information

		te			
		Total Memory Utilization	Implemented	Web Services	
Product	**Category**	**Feature/Attribute**	**Status**	**Protocol Used**	**Additional Information**
		swapRate.inRate	Implemented	Web Services	
		Swap Rate	Implemented	Web Services	
		Total Swap Rate	Implemented	Web Services	
		Schedule wait time	Implemented	Web Services	Waiting to be scheduled time in percent

Management APIs used by this data collector:

API	Protocol Used	Transport layer protocol used	Incoming ports used	Outgoing ports used	Supports authentication	Requires only 'Read-only' credentials	Supports Encryption	Firewall friendly (static ports)
VMware REST API	Web Services	HTTP/HTTPS	80/443		true	true	true	true

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